EFFECT OF ENTERPRENUERSHIP ON SELECTED ECONOMIC DEVELOPMENT INDICATORS IN NIGERIA (1986 - 2022)

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COLLEGE OF ARTS, MANAGEMENT AND SOCIAL SCIENCES

CHRISLAND UNIVERSITY, ABEOKUTA, OGUN STATE.

AUGUST, 2023

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BY

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**A RESEARCH PROJECT IN THE DEPARTMENT OF ECONOMICS SUBMITTED TO THE COLLEGE OF ART, MANAGEMENT AND SOCIAL SCIENCES.
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AUGUST, 2023

DECLARATION

I, George Favour Orire, do hereby declare that this project is entirely my work and composition. The work embodied in this project has not been submitted in candidature for any degree and is not concurrently being submitted for any other degree. All references made to works of other persons have been duly acknowledged.

…… ……. ……

Name of student Signature Date

CERTIFICATION

We certify that this research work was carried out by George Favour Orire in the Department of Economics, College of Art Management and Social Sciences, Chrisland University, Ogun State. The research work is considered adequate in partial fulfillment of the requirements for the award of the degree of Bachelor of Science (B.Sc) in Economics.

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DEDICATION

I dedicate this project to the almighty God, the giver of life, the author and finisher of my faith, the one who has kept me this far in life, may his name be highly praised.

I also dedicate this project to my parents, who has brought me to this world, and also to my siblings and friends who has supported me this far in life.

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LIST OF ABBREVATION

ENT: Entrepreneurship

SSE: Secondary school enrollment

GEH: Government expenditure on health sector

ECG: Economic Growth

INQ: Income Inequality

UR: Unemployment

CAP: Gross Fixed Capital Formation

SAP: Structural Adjustment Programme

GDP: Gross Domestic Product

ADRL: Autoregresive Distributed Lag

ADF: Augmented Dickey-Fuller

PP: Phillips-Perron

FDI: Foreign Direct Investment

NAPEP: National Poverty Eradication Program

NEEDS: National Economic Empowerment and Development Strategies

SURE-P: Subsidy Reinvestment and Empowerment Programme

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ABSTRACT

Entrepreneurship development is an indispensable factor in the reduction of poverty and income gap in developing countries where income inequality is high and economic growth is sluggish. Previous studies only focuses on the relationship between entrepreneurship and income inequality or entrepreneurship and economic growth, so this research aims to examine the relationship between entrepreneurship and all the determinants of economic development. This study analysed the trends of entrepreneurship, income inequality, unemployment and economic growth in Nigeria, with a view to examining the effect of entrepreneurship and investigating the direction of causality among entrepreneurship and income inequality, unemployment and economic growth in Nigeria. Ex-post facto research design was used. The study was anchored on Human Capital Theory by Theodore Schultz in 1960. Secondary data on entrepreneurship, income inequality, unemployment and economic growth from 1986 to 2022 were extracted from sources, namely, World Bank's Entrepreneurship Database (WBED), Standardized World Income Inequality Database (SWIID), and World Development Indicators (WDI). Data were subjected to descriptive and inferential analyses, using Ordinary Least Square (OLS) and Granger causality tests. The findings showed that there were fluctuations in entrepreneurship, income inequality, unemployment and economic growth, especially in 1990's due to political instability and changes in governments; from Ordinary Least Square revealed that entrepreneurship has significant negative effect on income inequality and unemployment; and significant positive effect on economic growth in Nigeria. Also, Granger Causality test showed a unidirectional causal relationship from entrepreneurship to unemployment and economic growth, but no causal relationship between entrepreneurship and income inequality in Nigeria. The study therefore concluded that there is a need to create an enabling environment for entrepreneurship to flourish and promote growth, ensure reduction in income inequality and unemployment in Nigeria.

Keyword: Entrepreneurship, Economic growth, Income inequality, Unemployment

CHAPTER ONE

INTRODUCTION

1.1 Background to the Study

Economic development is a term used to describe the process by which an economy's wealth, productivity, and standard of living improve over time leading to a rise in the level of the national income. It involves the use of various policies and strategies to encourage growth, increase economic activity, and enhance the overall well-being of a society. Economic development is taken to be the structural transformation of an economy by introducing more mechanized and updated technologies to increase labour productivity, employment, incomes, and standard of living of the population. Economic development should be accompanied by improvements in infrastructure, as well as social, political, and institutional factors to facilitate transformation of the economy (Myint & Krueger, 2016). Economic development is regarded as important for a country to reduce its poverty level by providing more employment, higher incomes, improved goods and services, and latest technologies of production. Economic development is defined as an increase in a country's wealth and standard of living (Charles Potters, 2021). For example, improved productivity, higher literacy rates, and better public education are all consequences of economic development in a country.

One of the key goals of economic development is to reduce poverty level and increase income levels, particularly in developing countries such as Nigeria. However, achieving this goal can be challenging, particularly in countries where income inequality is high and economic growth is sluggish. Income inequality, economic growth, poverty level and unemployment are critical factors as well as the significant determinant of economic development.

Economic growth could be seen as the process by which the productivity of an economy increases over a given period, leading to a rise in the level of the national income. When there is economic growth, it shows in the form of an increase in income level, an expansion in the labour force, an increase in the total capital stock of the country, and a higher volume of trade and consumption. According to Solow (1956), economic growth is the sustained increase in per capita output over time. This definition emphasizes the importance of long-term, sustained growth in the economy and the role of productivity in driving economic growth. It is the rate of change in the long-run aggregate production function of an economy, that is, the rate of change of its potential output or of its gross national product at constant prices, per head of population (Solow 1956). This definition emphasizes the role of productivity in driving economic growth and the importance of long-term, sustained growth.

Another important determinant of economic development is unemployment. The Nigerian economy needs to be more concerned about reduction in unemployment rate and level. It still remains one of the government's primary macroeconomic goals. A number of major developmental issues, including unemployment, are becoming more widespread in Nigeria. The main objective of the government and international organizations' policies is to lower the unemployment rate. Since the population explosion began, unemployment has been a feature of developing countries. Unemployment is considered to be a key measure of the health of the economy because it shows whether or not employees can find gainful employment and contribute to the economy's productive output (Solow 1956).

Piketty, (2014) defines income inequality as the concentration of income and wealth among a small share of the population, which can lead to social and economic instability. Income inequality has increased in many countries over the past few decades due to factors such as

globalization, technological change, and tax policies. Income inequality can negatively impact economic growth, social cohesion, and overall well-being. Excessive income inequality can lead to reduced social mobility, as individuals from lower-income backgrounds face barriers in accessing education, healthcare, and economic opportunities. This can result in a less skilled workforce and hinder overall economic development (Piketty, 2014). From an economic standpoint, income inequality can impact consumer spending patterns and aggregate demand. When a significant portion of the population has limited purchasing power, it can constrain economic growth by reducing demand for goods and services (Ostry, Berg & Tsangarides, 2014).

Entrepreneurship has been widely recognized as a potential solution to addressing unemployment, income inequality, and fostering economic growth. Entrepreneurial activities contribute to job creation, income generation, and innovation, which in turn can have positive effects on these economic indicators. Entrepreneurship can help combat unemployment by creating new job opportunities. Startups and small businesses often hire locally, leading to increased employment rates. When individuals start their own businesses or enterprises, they can generate employment for themselves and others. Entrepreneurial ventures have the potential to reduce income inequality by generating wealth and redistributing it throughout society. Small business ownership allows individuals from diverse backgrounds to participate in economic activities and increase their income. Entrepreneurship stimulates economic growth by driving innovation, competition, and productivity improvements. New businesses introduce innovative products, services, and processes, which can lead to increased consumer demand and economic expansion (Acs & Audretsch, 1990).

1.2 Statement of the problem

Recent challenging problems relating to economic development in Nigeria have hindered the country's progress and created barriers to achieving sustainable growth. These problems include a significant infrastructure deficit, high unemployment rates, corruption and governance issues, inadequate access to finance, inflation and exchange rate volatility, security challenges, weak agricultural sector, and persistent inequality and poverty. These challenges pose substantial obstacles to economic diversification, investment attraction, job creation, and equitable development.

Nigeria faces significant income inequality, with a substantial wealth gap and uneven distribution of resources among its population. This income disparity undermines social cohesion, perpetuates poverty rates, and limits opportunities for upward mobility. The concentration of wealth among a few individuals or groups exacerbates social tensions, leading to social unrest and hampering the overall well-being of the society (Cingano, 2014). Addressing the issue of income inequality is essential to promote social justice, reduce poverty, and foster inclusive economic growth in Nigeria. Nigeria faces sluggish economic growth, marked by limited diversification, over-reliance on oil exports, and inadequate job creation. The lack of sufficient employment opportunities contributes to high unemployment rates, particularly among the youth population, leading to social unrest and a loss of human capital potential. Addressing these interconnected challenges is crucial for fostering equitable economic growth, reducing income inequality, and providing opportunities for sustainable livelihoods in Nigeria.

Nigeria faces significant challenges in infrastructure development, including inadequate transportation networks, power supply deficiencies, and limited access to clean water and sanitation facilities. The lack of quality infrastructure hampers productivity, increases business

costs, and impedes economic growth (Shehu & Nazim, 2022) However, Nigeria experiences high unemployment rates, particularly among the youth population. Insufficient job creation, limited access to quality education and vocational training, and a growing labour force exacerbate the unemployment problem. This poses social and economic challenges, including income inequality, poverty, and social unrest.

Nigeria is often referred to as the giant of Africa is blessed with abundant natural resources including vast arable land for agricultural activities. The agricultural sector has historically played a crucial role in the Nigerian economy employing a significant portion of the population. Despite the agricultural sector's potential, Nigeria faces challenges such as limited access to credit, outdated farming techniques, inadequate infrastructure, and post-harvest losses. These issues impede agricultural productivity, food security, and rural development. Nigeria can unlock this potential of the agricultural sector solve unemployment problem, stimulate economic growth, achieve sustainable economic development and alleviate poverty among its population (Rome, 2017).

Nigeria's economy heavily relies on oil exports, which accounts for a significant portion of the country’s GDP and government revenue, making it vulnerable to fluctuations in global oil prices. The volatility in oil prices and the COVID-19 pandemic have exposed the country to significant revenue shocks, impacting government finances and hindering economic growth. The Nigerian economy remains largely dependent on the oil sector, neglecting the development of other key sectors such as agriculture, manufacturing, and services. This lack of diversification leaves the economy susceptible to external shocks and inhibits the creation of employment opportunities and inclusive growth (Yu Li & Umair, 2023).

Entrepreneurship, defined as the process of creating and scaling innovative business ventures, has the potential to contribute positively to income distribution, economic growth, and employment generation (Shane & Venkataraman, 2000). Entrepreneurship promotes job creation by encouraging individuals to start and expand businesses. Entrepreneurs establish new ventures, leading to the creation of employment opportunities for others. Through the growth of small and medium-sized enterprises (SMEs), entrepreneurship can generate a significant number of jobs across various sectors, reducing unemployment rates and providing livelihood opportunities. Entrepreneurial ventures drive economic growth and diversification by introducing new products, services, and business models. By identifying market gaps and addressing unmet needs, entrepreneurs stimulate innovation, increase productivity, and contribute to the expansion of the economy beyond traditional sectors. Entrepreneurship enhances economic resilience and reduces dependence on a single industry, fostering a more sustainable and inclusive economic growth trajectory (OECD 1996).

Entrepreneurship can play a role in reducing income inequality by creating income-generating opportunities for individuals from diverse backgrounds. By fostering entrepreneurship among marginalized groups and promoting inclusivity, entrepreneurial activities enable individuals to generate income, accumulate wealth, and contribute,,,, to a more equitable distribution of resources. Access to entrepreneurship also empowers individuals to break free from the cycle of poverty, fostering social mobility and reducing income disparities.

By promoting entrepreneurship as a solution to income inequality, economic growth, and unemployment, Nigeria can unlock the potential of its population, spur innovation, and drive sustainable development. Investing in entrepreneurship development programs, providing access

to finance, and fostering an enabling ecosystem will help create a conducive environment for entrepreneurial activities, resulting in inclusive economic growth and reduced income disparities.

Empirically, evidence obtained from various studies within and outside Nigeria both at country-specific and cross-country level indicate that there is relationship between entrepreneurship and income inequality. For instance, Acs and Szerb (2009), Caliendo, Fossen and Kritikos, (2009), Guzman and Stern, (2016), Krieger, Fritsch, and Sorgner, (2008) investigated the relationship between entrepreneurship and income inequality. They found that higher levels of entrepreneurship were associated with lower income inequality. They argued that entrepreneurship provides opportunities for wealth creation and social mobility, thus contributing to more equal income distribution. They suggested that this relationship could be attributed to the fact that successful entrepreneurs accumulate substantial wealth, leading to increased income disparities. They found that self-employed individuals had higher levels of income inequality compared to wage earners.

Moreover, a large body of evidence from the literature has argued that entrepreneurship has impact on economic growth. Braunerhjelm et al (2010), Autio et al (2013) and Stam, Stam, and Audretsch (2014). They found a positive relationship between entrepreneurial activity and economic growth. They found a positive association between entrepreneurship and economic growth, indicating that entrepreneurial activity contributed to higher levels of economic expansion. Their findings indicated that entrepreneurship played a crucial role in knowledge diffusion and innovation, leading to higher levels of economic growth. Moreover, they highlighted the importance of supportive policies and institutional factors in fostering entrepreneurship and driving economic expansion.

However, the studies that examined the relationship between entrepreneurship and economic development are Acs, Desai and Hessels, (2008), Wennekers and Thurik, (1999), Acs, Stam, Audretsch and O'Connor, (2017), Haltiwanger et al (2013). They found a positive relationship between entrepreneurship and economic development, suggesting that higher levels of entrepreneurship contribute to economic growth and development and that entrepreneurship plays a significant role in driving economic development.

Moreover, there are studies that examined the impact of entrepreneurship on reducing unemployment. For example, Audretsch and Keilbach, (2004). The study found that entrepreneurship had a positive impact on employment growth, indicating that higher levels of entrepreneurial activity were associated with lower unemployment rate. The study found that young and small firms, which are often associated with entrepreneurial activity, made significant contributions to net job creation. This suggests that entrepreneurship can have a positive impact on reducing unemployment by creating new job opportunities. Entrepreneurship can serve as an alternative pathway for individuals facing unemployment.

Empirically, evidence obtained from various studies within and outside Nigeria both at country-specific and cross-country level examines the relationship between entrepreneurship and the determinants of economic development. Yusuf and Ibrahim, (2020) examined the impact of entrepreneurship on unemployment in Nigeria, focusing on the long-run and short-run dynamics for the period of 1990 to 2019. The study finds a negative relationship between entrepreneurship and unemployment in Nigeria in both the long run and short run. An increase in entrepreneurial activities leads to a reduction in the unemployment rate. The study also reveals that entrepreneurship has a stronger impact on unemployment in the long run compared to the short run.

According to Acs and Armington (2006), entrepreneurship has the potential to reduce income inequality by creating employment opportunities for the poor and reducing their dependence on wages. Similarly, Berggren and Jordahl (2006) argue that entrepreneurship can reduce income inequality by increasing the share of income earned by the self-employed.

Abubakar and Yusuf (2020) examined the relationship between entrepreneurship, economic development, and p,overty alleviation in Nigeria. finds a positive and significant relationship between entrepreneurship and economic development. Adekoya and Osabohien (2019) the study finds a positive and significant relationship between entrepreneurship and economic growth in Nigeria.

There are potential gaps in the research that I observed and future researchers could explore. Most of the study focuses on reviewing empirical evidence on the relationship between entrepreneurship and determinant of economic development, which are income inequality, economic growth, and unemployment. However, establishing causal relationships can be challenging due to endogeneity issues. Future research could employ more rigorous methodologies, such as instrumental variable approaches or natural experiments, to address endogeneity concerns and identify more robust causal effects.

1.3 Research Questions

Answers to the followings questions are expected to give insight on the effects of entrepreneurship on income inequality, unemployment and economic growth in Nigeria.

1. Does entrepreneurship influence income inequality, unemployment and economic growth in Nigeria?

2. What is the effect of entrepreneurship on the income inequality, unemployment and economic growth in Nigeria?

1.4 Objectives of the Study

The broad objective of the study is to investigate the relationship between entrepreneurship and the determinant of economic development in Nigeria. Therefore, the specific objectives are to:

i. analyse the trends of entrepreneurship, income inequality, unemployment and economic growth in Nigeria
ii. examine the effect of entrepreneurship on income inequality, unemployment and economic growth in Nigeria.
iii. investigate the direction of causality among entrepreneurship and income inequality, unemployment and economic growth in Nigeria

1.5 Research Hypothesis

For the purpose of this study, it is important to set hypothesis in which the objectives to be achieved will be tested and relevant conclusion can be drawn.

Hypothesis I

H0: Entrepreneurship does not have significant effect on the determinant of economic development in Nigeria.

H1: Entrepreneurship has significant effect on the determinant of economic development in Nigeria.

Hypothesis II

H0: There is no causal relationship between entrepreneurship and the determinants of economic development in Nigeria.

H1: There is causal relationship between entrepreneurship and the determinants of economic development in Nigeria

1.6 Scope of the Study

This study examined the relationship among entrepreneurship, income inequality, economic growth and unemployment in Nigeria for the periods of 1986 to 2022. The selected time frame from 1986 to 2022 encompasses a significant period of Nigeria's economic history, including the implementation of the Structural Adjustment Programme (SAP). The SAP was a comprehensive set of economic reforms implemented in Nigeria from the mid-1980s to the early 1990s, with the aim of restructuring the economy, promoting market liberalization, and addressing economic imbalances. By examining the period from 1986 to 2022, this study seeks to analyse the effects of entrepreneurship on income inequality, economic growth, and unemployment, taking into account the context and dynamics of Nigeria's economic reforms during the SAP era. This period allows for a comprehensive understanding of the relationship between entrepreneurship and the determinants of economic development, as it encompasses pre-SAP, SAP implementation, and post-SAP periods, providing insights into the changes and trends that have unfolded over time (Nduka, Chukwu & Ugbor, 2013).

1.7 Significance of the Study

Nigeria is grappling with persistent economic challenges, including income inequality and high unemployment rates, despite its potential for economic growth. This study aims to shed light on the role of entrepreneurship as a potential solution to these challenges. Understanding how

entrepreneurship can impact income distribution, create jobs, and drive economic growth is crucial for formulating targeted policies and interventions that can foster inclusive and sustainable development.

The study's findings can provide valuable insights for Nigeria's economic recovery and resilience, especially in the aftermath of global events like the COVID-19 pandemic. Entrepreneurship has the potential to spur innovation, diversify the economy, and create new opportunities, contributing to a more robust and resilient economic ecosystem. This study's outcomes can serve as evidence-based inputs for policy formulation and decision-making, guiding the design of initiatives that promote entrepreneurship, support startups, and enhance access to finance and resources for aspiring entrepreneurs (Albert, Caggese, Gonzalez & Victor, 2023). The study on the effect of entrepreneurship on income inequality, economic growth, and unemployment in Nigeria offers several benefits. One of the benefit of this study is that policymakers and stakeholders can use the empirical evidence generated from this study to make informed decisions on entrepreneurship-related policies, ensuring that initiatives are well-targeted and have a higher chance of success.

Unemployment, particularly among the youth, is a pressing issue in Nigeria. Understanding how entrepreneurship can be leveraged to create self-employment opportunities and generate jobs for others can empower the youth to become drivers of economic growth rather than victims of unemployment. We are in a country where universities turn out thousands of student every year into the labour market to seek employment and they all cannot find jobs, entrepreneurship can be a solution in this instance to unemployment (Oyelola, Igwe & Peluola, 2014)

The study can also contribute to the existing body of academic literature on entrepreneurship, income inequality, economic growth, and unemployment in the context of Nigeria. It can pave

the way for further research and discussions on the intricate relationship between entrepreneurship and the determinants of economic development. By understanding the dynamics of entrepreneurship and its impact on the determinants of economic development, the study can provide actionable recommendations to promote entrepreneurship as a catalyst for positive change in Nigeria's economy. Embarking on research on the effect of entrepreneurship on income inequality, inflation, economic growth, and unemployment in Nigeria can provide valuable insights for policymakers, researchers, entrepreneurs, development organizations, society at large and other beneficiaries.

1.8 DEFINITION OF TERMS

Entrepreneurship:

Entrepreneurship refers to the process of identifying and pursuing opportunities to create, innovate, and manage a business venture. It involves taking risks, organizing resources, and applying innovative ideas to create value, generate profits, and drive economic growth (Shane & Venkataraman, 2000).

Income Inequality:

Income inequality refers to the unequal distribution of income among individuals or households within a society. It measures the disparities in income levels and the concentration of wealth, often represented by indicators such as the Gini coefficient or the ratio of the highest to the lowest income earners (Atkinson, 2015).

Economic Growth:

Economic growth refers to the increase in the production of goods and services in an economy over time. It is typically measured by changes in real gross domestic product (GDP) and indicates the expansion of economic activity, increased productivity, and improved living standards (Mankiw, Romer, & Weil, 1992).

Unemployment:

Unemployment refers to the situation in which individuals of working age who are actively seeking employment are unable to find jobs. It represents the proportion of the labour force that is without work but available for and actively seeking employment (Blanchard & Fischer, 1989)

CHAPTER TWO

LITERATURE REVIEW

This chapter presents a review of the relevant literature on the effect of entrepreneurship on income inequality, economic growth, and unemployment in Nigeria. The chapter is divided into three sections, the first section focuses on the conceptual review. The second section discusses the theoretical review, while the third section presents the detailed empirical review of literature in Nigeria, developing and developed countries.

2.1 Conceptual Review

The conceptual review involves an extensive examination on effects of entrepreneurship on the determinants of economic development in Nigeria. The conceptual review, also attempts to provide a detailed discussion on the various variables used in this study.

2.1.1 Income Inequality

Income inequality refers to the unequal distribution of income among individuals or households in a particular society or country. It is often measured by the Gini coefficient, which ranges from 0 (perfect equality, where everyone has the same income) to 1 (perfect inequality, where one person or group has all the income and everyone else has none. There are many factors that can contribute to income inequality, including differences in education, skills, and job opportunities; discrimination based on gender, race, or ethnicity; and government policies related to taxation, social welfare programs, and labour laws. Some scholars argue that income inequality is not necessarily a bad thing, as it can provide incentives for people to work hard and innovate, and can also fund public goods and services through taxation (Mankiw, 2014).

According to Atkinson (2015), income inequality refers to the unequal distribution of income among individuals or households in a given society. It is typically measured using a Gini coefficient, which ranges from 0 (perfect equality) to 1 (perfect inequality).

Piketty, (2014) defines income inequality as the concentration of income and wealth among a small share of the population, which can lead to social and economic instability. He argues that income inequality has increased in many countries over the past few decades due to factors such as globalization, technological change, and tax policies. Saez and Zucman (2016) define income inequality as the share of total income that accrues to the top 1% of earners in a society. They argue that this measure provides a more accurate picture of income inequality than other measures, such as the Gini coefficient. Milanovic (2016) defines income inequality as the extent to which income is distributed unevenly among individuals or households in a society. He argues that income inequality can have both positive and negative effects on economic growth and development. Stiglitz (2012) defines income inequality as a situation where some individuals or groups in society have significantly more income or wealth than others, which can lead to social and economic instability. He argues that income inequality is a key challenge facing many countries today and that policies are needed to address it.

2.1.2 Unemployment

It is a macroeconomic problem that affects the economy as a whole and has various social and economic consequences. According to Blanchard & Johnson (2013), unemployment is "the situation in which people who are willing and able to work cannot find jobs." This definition emphasizes the importance of willingness and ability to work in defining unemployment. Unemployment is a situation in which persons capable and willing to work are unable to find a suitable paid job at a given wage rate at a particular period of time. Unemployment is a state of

affairs where a person actively seeks employment but is unsuccessful in doing so. One important indicator of the economy's health is unemployment. The unemployment rate is the indicator of unemployment that is most frequently used. It is determined by dividing the total number of employed persons by the total number of unemployed people. To certain unemployed people who meet the eligibility requirements, many governments offer unemployment insurance. (Mankiw, 2021).

Marx, (1867) emphasizes the structural factors that contribute to unemployment, such as the exploitation of labour under capitalism. In the words of Keynes (1936), unemployment is "the involuntary idleness of a substantial proportion of the working population." This definition emphasizes the involuntary nature of unemployment and the impact it can have on individuals and the economy. Phelps (1968) defines unemployment as "the excess supply of labour at the prevailing wage rate." This definition emphasizes the role of wages in determining the level of unemployment in an economy. In the words of Friedman (1968), unemployment is "a situation in which a person is willing to work at the going wage rate but cannot find a job." This definition emphasizes the willingness of individuals to work and the importance of labour market dynamics in determining unemployment. Gordon (1990) describes unemployment as "the failure of the economy to generate a sufficient number of jobs for all those who are willing and able to work." This definition emphasizes the impact of macroeconomic factors on unemployment, such as the level of aggregate demand in the economy.

2.1.3 Entrepreneurship

Entrepreneurship is the process of identifying, creating, and pursuing opportunities for innovative business ventures. It involves taking calculated risks, being resourceful, and adapting to changing market conditions. Entrepreneurs are individuals who take the initiative to develop

new ideas and concepts and bring them to fruition. Entrepreneurship is not limited to starting a new business; it can also involve introducing new products or services, improving existing processes or products, and creating social enterprises. It plays a crucial role in economic development and job creation, as new businesses often lead to the creation of new jobs and the stimulation of local economies.

The field of entrepreneurship has received significant attention in recent years, with numerous academic studies and theories aimed at understanding the entrepreneurial process. Scholars have identified various factors that contribute to entrepreneurial success, including access to capital, human resources, networks, and support systems. Entrepreneurship is a complex and dynamic phenomenon, influenced by a range of personal, social, and economic factors. Successful entrepreneurs possess a combination of skills, knowledge, and personal traits, including creativity, perseverance, risk-taking, and adaptability. With the right combination of resources and opportunities, anyone can become an entrepreneur and make a significant impact on the world.

According to Schumpeter (1934), entrepreneurship is "the process of creative destruction by which old ways of doing things are replaced by new ways." This definition emphasizes the role of entrepreneurs in disrupting existing markets and creating new ones through innovation. Shane & Venkataraman (2000) define entrepreneurship as "the discovery, evaluation, and exploitation of opportunities." This definition emphasizes the fact that entrepreneurship involves identifying opportunities in the market and taking action to capitalize on them. Stevenson and Gumpert (1985) describe entrepreneurship as "the pursuit of opportunity beyond the resources currently controlled." This definition emphasizes the fact that entrepreneurship involves taking calculated risks to pursue new opportunities.

Hisrich, Peters and Shepherd (2017) define entrepreneurship as "the process of creating value by bringing together a unique combination of resources to exploit an opportunity." This definition emphasizes the fact that entrepreneurship involves identifying and leveraging unique resources to create value in the market. Kuratko (2016) describes entrepreneurship as "the dynamic process of creating incremental wealth by individuals or teams who identify opportunities, allocate resources, and create value." This definition emphasizes the fact that entrepreneurship involves not only identifying opportunities, but also effectively managing resources and creating value for customers.

In conclusion, entrepreneurship is a term that refers to the process of creating, managing, and growing a new business venture. These different authors offer different definitions of entrepreneurship, each emphasizing different aspects of the phenomenon. Some focus on the innovative nature of entrepreneurship, while others emphasize the importance of resourcefulness, risk-taking, and opportunity identification in entrepreneurial activity.

2.1.4 Economic Growth

Economic growth can be defined as the growth rate of per capita GDP. This definition emphasizes the role of output per person and technological progress in driving economic growth and measuring economic growth. The following are definitions of inflation by authors and economists. Barro and Sala-i-Martin (1995) describe economic growth as the increase in the value of goods and services produced by an economy over time, measured by the growth rate of real GDP. Acemoglu and Robinson (2012) define economic growth as the process whereby an economy's potential output is steadily increasing over time, giving rise to a sustained expansion in the standard of living.

According to Solow (1956), economic growth is the sustained increase in per capita output over time. This definition emphasizes the importance of long-term, sustained growth in the economy and the role of productivity in driving economic growth. It is the rate of change in the long-run aggregate production function of an economy, that is, the rate of change of its potential output or of its gross national product at constant prices, per head of population. This definition emphasizes the role of productivity in driving economic growth and the importance of long-term, sustained growth. According to Edward Denison (1962), economic growth is the increase in the capacity of an economy to provide goods and services, compared from one period of time to another. It emphasizes the role of increasing output in driving economic growth and the importance of comparing growth over time.

According to Edward Denison (1962), economic growth is the increase in the capacity of an economy to provide goods and services, compared from one period of time to another. According to Douglass North and Robert Thomas (1973), economic growth is a process of structural change whereby the economies of individual countries or groups of countries become more differentiated and integrated into the world economy.

2.2 Theoretical Review

2.2.1 Schumpeterian Theory of Economic Development

The Schumpeterian theory of economic development is a theory of economic growth that emphasizes the role of innovation and entrepreneurship. It was developed by the Austrian economist Joseph Schumpeter in his book The Theory of Economic Development (1911). This theory emphasizes the disruptive nature of entrepreneurial activities and the importance of innovation in fostering economic progress. According to Schumpeter, entrepreneurs are agents of

change who introduce new products, processes, technologies, and business models into the market. Schumpeter believed that entrepreneurship was driven by innovation, which he defined as the "commercial or industrial application of something new" (Schumpeter, 1942). This could involve the introduction of a new product or service, a new production process, a new market or distribution channel, or even a new organizational structure.

Schumpeter argued that this process of innovation and creative destruction was essential for economic growth, as it led to increased productivity, efficiency, and competitiveness. However, he also acknowledged that this process could lead to income inequality, as the benefits of innovation were often concentrated among a small group of entrepreneurs and investors. Creative destruction refers to the continuous process of innovation, where the introduction of new products and technologies replaces outdated ones. This process not only leads to economic growth but also involves the destruction of existing industries, technologies, and employment patterns. Schumpeter argued that this process of creative destruction is necessary for sustained economic development. Schumpeter identified entrepreneurial profit as a key driver of economic development. Entrepreneurs generate profit by introducing innovative products or processes that offer unique value to customers. The pursuit of profit incentivizes entrepreneurs to take risks, invest in research and development, and seek out new market opportunities.

The Schumpeterian theory of economic development can have mixed effects on income inequality. While entrepreneurship and innovation can contribute to economic growth and upward mobility, they can also lead to income disparities, especially in the short term (Gompers & Lerner, 2001).

According to Schumpeter, entrepreneurs play a crucial role in driving economic development through innovation and creative destruction. Successful entrepreneurs who introduce disruptive

technologies or business models can reap substantial rewards. This can lead to an increase in income inequality, as highly successful entrepreneurs accumulate significant wealth, while others may struggle to adapt or face displacement due to creative destruction. As a result, income disparities can widen as a small group of successful entrepreneurs capture a larger share of the economic gains.

The Schumpeterian theory emphasizes the importance of technological progress in economic development (Schumpeter, 1911). Innovation and the adoption of new technologies can lead to increased productivity and economic growth. However, technological progress may also have a dual impact on income inequality. In the short term, it can contribute to job displacement and wage polarization, as certain occupations become obsolete while others requiring higher skills or specialization command higher wages. This can exacerbate income inequality if workers in declining industries face challenges in transitioning to new sectors. In the long term, technological progress can lead to overall economic growth, creating new opportunities for income generation and potentially reducing inequality.

The theory does not explicitly aim to solve income inequality. Instead, it focuses on understanding the role of entrepreneurship and innovation in driving economic growth and development. However, the theory offers insights that can inform policy approaches to address income inequality

2.2.2 Human Capital Theory

The human capital theory of economic development is a concept that focuses on the role of education, skills, and knowledge as essential factors in promoting economic growth and development. It posits that investing in human capital, through education and training, contributes to higher productivity, innovation, and overall economic progress (Schultz, 1961).

The roots of the human capital theory can be traced back to classical economic thinkers such as Adam Smith and Alfred Marshall. They recognized the importance of education and skills in enhancing labour productivity and argued that human capital is a valuable economic asset. However, their ideas were not developed into a comprehensive theory at the time. The theory gained prominence with the work of Theodore W. Schultz, an American economist who received the Nobel Prize in Economic Sciences in 1979 for his contributions to the theory. He argued that investments in education and training were a form of human capital, which could lead to increased productivity, higher wages, and better economic outcomes for individuals and society as a whole (Schultz, 1961).

The theory seeks to address the problem of understanding the relationship between education, skills, and economic outcomes. It aims to explain how investments in human capital, such as education and training, contribute to economic growth, productivity, and individual earnings (Psacharopoulos, 1994). The problem is to understand how investments in human capital translate into higher productivity and overall economic growth. Human capital refers to the knowledge, skills, and abilities that individuals acquire through education and training, and the theory seeks to explain how these investments enhance productivity at the individual, organizational, and national levels. The theory also addresses the problem of explaining the relationship between human capital and individual earnings. It seeks to understand how

education, skills, and other forms of human capital investment influence an individual's ability to secure higher-paying jobs, attain promotions, and increase their income over time.

Another problem that the theory seeks to solve is the issue of skill mismatches and unemployment. It explores how disparities between the skills possessed by workers and the skills demanded by employers affect unemployment rates and labour market efficiency.

With the rapid advancement of technology, the Human Capital Theory seeks to understand how individuals and economies can adapt to technological change. It explores how investments in human capital, particularly in acquiring digital literacy, cognitive flexibility, and adaptability, can facilitate adjustment to changing job requirements and technological advancements. The theory informs strategies for lifelong learning, upskilling, and reskilling to ensure individuals and societies remain competitive in the evolving labour market.

Finally, the theory addresses the broader problem of economic development and sustainable growth. It contributes to economic development, policymakers can formulate strategies to promote sustainable economic growth, reduce poverty, and improve living standards.

Despite its broad applications, the human capital theory has also been subject to criticism and debate, particularly with regards to its neglect of social and cultural factors that shape educational and occupational outcomes. Nonetheless, the theory has influenced policy debates and led to the development of policies and programs that seek to promote investments in human capital, such as education and training programs, and labour market policies that encourage skills development and lifelong learning (Mincer, 1974)

Overall, the theory seeks to solve the problem of understanding the relationship between education, skills, and economic outcomes. By providing insights into the mechanisms through

which human capital investments influence productivity, earnings, inequality, and economic development, the theory helps guide policy interventions and educational reforms to optimize human capital development and enhance economic outcomes.

2.2.3 Phillips Curve Theory

The Phillips curve theory is named after the economist A.W. Phillips, who first observed a negative relationship between unemployment and wage inflation in the United Kingdom in the 1950s. His work was based on an analysis of data on unemployment and wage changes in the UK over the period 1861-1957. Phillips observed that when unemployment was low, wages tended to rise more quickly, and when unemployment was high, wages tended to rise more slowly. This led him to suggest that there was a trade-off between unemployment and inflation, with low unemployment leading to higher inflation and high unemployment leading to lower inflation.

Phillips's work gained widespread attention among economists and policymakers, and his observations were later formalized into the Phillips curve theory. The Phillips curve became a central concept in macroeconomic analysis and policymaking during the 1960s and 1970s, as governments sought to balance the goals of low unemployment and stable prices.

However, the relationship between unemployment and inflation became less clear in the 1970s, as high inflation and high unemployment coexisted in a phenomenon known as stagflation. This challenged the assumptions underlying the Phillips curve theory and led to the development of new models that incorporated expectations of future inflation and other factors that influenced the relationship between unemployment and inflation (Phillips, 1958).

One of the most notable criticisms of the Phillips curve is the so-called "Phillips curve tradeoff," which suggests that there is a tradeoff between inflation and unemployment in the short run. However, this tradeoff is subject to challenge, as the relationship between inflation and unemployment can change over time and may not always be negative.

Despite these challenges, the Phillips curve remains a useful tool for policymakers in understanding the relationship between unemployment and inflation and guiding macroeconomic policy. It has also been subject to ongoing refinement and development, as economists continue to seek a better understanding of the complex dynamics of the macroeconomy.

2.3 Empirical Review

2.3.1 Entrepreneurship and Economic Growth

The relationship between entrepreneurship and economic growth is a well-established one. Entrepreneurship has long been recognized as a driver of economic growth, as it creates new businesses, products, and services that contribute to the development of new industries and the expansion of existing ones. Agu and Eze (2021) examined the relationship between entrepreneurship and economic growth in Nigeria, focusing on understanding the impact of entrepreneurship on key indicators of economic growth for the period of 1990 to 2020. The authors employ a longitudinal analysis, utilizing panel data techniques such as fixed-effects regression models and dynamic panel estimations. The study finds a positive and significant relationship between entrepreneurship and economic growth in Nigeria. The study concludes that entrepreneurship plays a crucial role in driving economic growth in Nigeria. Policymakers should prioritize creating an enabling environment that fosters entrepreneurship by providing

access to finance, promoting entrepreneurship education and training, and implementing supportive policies.

Adeniyi and Adewale (2020) examined the relationship between entrepreneurship, economic growth, and productivity in Nigeria, with a focus on understanding the mechanisms through which entrepreneurship influences economic performance for the period of 1990 to 2019. The authors employ time series analysis, using the Autoregressive Distributed Lag (ARDL) model to analyse the long-run and short-run relationships between entrepreneurship, economic growth, and productivity. They include variables such as entrepreneurship index, GDP growth rate, labour productivity, investment, and innovation indicators. The analysis also considers control variables, such as human capital and infrastructure development, to account for other factors that may influence the relationship. The study finds a positive and significant relationship between entrepreneurship, economic growth, and productivity in Nigeria. It reveals that entrepreneurship positively impacts economic growth by promoting innovation, investment, and job creation. The study also highlights the mediating role of productivity, indicating that entrepreneurship enhances productivity levels, leading to higher economic output and sustained economic growth.

Adeyemo and Adeyemi (2020) investigated the impact of entrepreneurship on economic growth in Nigeria for the period of 1981 to 2018. The study employs the autoregressive distributed lag (ARDL) bounds testing approach to examine the long-run and short-run relationships among the variables. The study finds a positive relationship between entrepreneurship and economic growth, while there is an inverse relationship between entrepreneurship and inflation.

Aremu, Adeyemo, and Olarewaju, 2019 investigated the relationship between entrepreneurship and economic growth in Nigeria for the period of 1981-2017. The study employs the autoregressive distributed lag (ARDL) bounds testing approach to examine the long-run and

short-run relationships among the variables. The study finds a positive and significant relationship between entrepreneurship and economic growth in Nigeria.

Adekoya and Osabohien (2019) investigated the relationship between entrepreneurship and economic growth in Nigeria for the period of 1995 to 2017. The study employs the Autoregressive Distributed Lag (ARDL) model to estimate the long-run and short-run relationships among the variables. The study finds a positive and significant relationship between entrepreneurship and economic growth in Nigeria. Adejumo (2017) investigated the impact of entrepreneurship on economic growth in Nigeria for the period of 1955 to 2017. The study employs the Autoregressive Distributed Lag (ARDL) model to estimate the long-run and short-run relationships among the variables. The study finds that entrepreneurship has positive and significant impact on economic growth.

Njoku and Isihak (2019) investigated relationship between entrepreneurship, income inequality, and economic growth in African countries for the period of 2005-2015. The study employs the Pooled Ordinary Least Squares (POLS). The study finds that negative relationship between entrepreneurship and income inequality, indicating that entrepreneurship may contribute to reducing income inequality in African countries.

Ismail and Soile (2019) examined the relationship between entrepreneurship, economic growth, and unemployment in Nigeria for the period of 1986-2016. The authors employ time series analysis, using the Autoregressive Distributed Lag (ARDL) model to analyse the long-run and short-run relationships between entrepreneurship, economic growth, and unemployment. The study finds a significant long-run relationship between entrepreneurship and economic growth, indicating that entrepreneurship contributes to economic development. Additionally, it finds a

negative relationship between entrepreneurship and unemployment, suggesting that entrepreneurship can help reduce unemployment in Nigeria.

Babalola and Falana (2020) examined the relationship between entrepreneurship, economic growth, and unemployment in Nigeria for the period of 1980-2016. The authors employ time series analysis, using the Autoregressive Distributed Lag (ARDL) model to analyse the long-run and short-run relationships between entrepreneurship, economic growth and economic development. The study finds a positive and significant relationship between entrepreneurship and economic growth, indicating that entrepreneurship promotes economic development in Nigeria.

Adeleke (2018) investigated the relationship between entrepreneurship and economic growth in Nigeria. The study covers the period from 1980-2017. The study employs the Autoregressive Distributed Lag (ARDL) model to estimate the long-run and short-run relationships among the variable. The study finds a positive and significant relationship between entrepreneurship and inclusive growth, indicating that entrepreneurship contributes to inclusive economic development in Nigeria.

Ogunsanya, Shoyinka and Adegbie (2020) examined relationship between entrepreneurship and economic growth for the period 1981-2018. The study employs the Autoregressive Distributed Lag (ARDL) model to estimate the long-run and short-run relationships among the variable. The study finds a positive relationship between entrepreneurship and economic growth, indicating that entrepreneurship plays a significant role in promoting economic development in Nigeria. Additionally, it finds a positive relationship between trade openness and economic growth, suggesting that increased trade can stimulate economic growth in the country.

Olokundun and Moses (2020) investigated the relationship between entrepreneurship, inflation, and economic growth in Nigeria for the period 1981-2016. The variables employed in the study are entrepreneurship, inflation rate, GDP growth rate, foreign direct investment (FDI), and trade openness. The study employs the Autoregressive Distributed Lag (ARDL) model to estimate the long-run and short-run relationships among the variable. The study finds a positive relationship between entrepreneurship and economic growth, while inflation has a negative effect on economic growth in Nigeria.

2.3.2 Entrepreneurship and Income inequality

Entrepreneurship can play a role in reducing income inequality by creating income-generating opportunities for individuals from diverse backgrounds. By fostering entrepreneurship among marginalized groups and promoting inclusivity, entrepreneurial activities enable individuals to generate income, accumulate wealth, and contribute to a more equitable distribution of resources. Access to entrepreneurship also empowers individuals to break free from the cycle of poverty, fostering social mobility and reducing income disparities. Several empirical studies have provided evidence for the positive relationship

Oladele and Oladele (2019) investigated the relationship between entrepreneurship and income inequality in Nigeria for the period 2000 – 2016. The study employs the autoregressive distributed lag (ARDL) bounds testing approach to examine the long-run and short-run relationships among the variables. The study finds that entrepreneurship has a positive and significant impact on income inequality in Nigeria. The study fills the gap in the literature by investigating the relationship between entrepreneurship and income inequality in Nigeria

Adegbite and Osabohien (2021) examined the impact of entrepreneurship on income inequality in Nigeria for the period 1981-2018. The study employs the Autoregressive Distributed Lag (ARDL) model to estimate the long-run and short-run relationships among the variables. The study finds a significant negative relationship between entrepreneurship and income inequality in the long-run, indicating that entrepreneurship can reduce income inequality in Nigeria. The study fills the gap in the literature by examining the impact of entrepreneurship on income inequality in Nigeria.

Anyanwu and Erhijakpor (2014) analysed the impact of structural change on income inequality and poverty reduction in Nigeria. This study covered the period 1980-2010. The variables employed are structural change (measured by sectoral share of employment), income inequality (measured by Gini coefficient), and poverty rate. The study employs the Autoregressive Distributed Lag (ARDL) model to estimate the long-run and short-run relationships among the variables. The study finds that structural change has a negative effect on income inequality and poverty reduction in Nigeria.

Tamvada (2010) examined the relationship between income inequality and entrepreneurship. This study covered the period 1990-2005. The study employs the Cross-country panel data analysis to estimate the long-run and short-run relationships among the variables. The study found that income inequality is positively correlated with the total number of entrepreneurs, but not with tax-declaring startups or high-growth entrepreneurs. This suggests that entrepreneurship can help to reduce income inequality, but only if it is inclusive and creates opportunities for people from all income levels.

Eriksson and Henrekson (2017) examined the relationship between entrepreneurship and income inequality in Sweden. This study covered the period 1990-2012. They use econometric

techniques such as regression analysis, employing panel data models to examine the relationship between entrepreneurship and income inequality. The study found that entrepreneurship is negatively correlated with income inequality in Sweden. This suggests that entrepreneurship can help to reduce income inequality in countries with a strong social safety net and a supportive institutional environment.

Eggers, Kraus and Chang (2017) examined the relationship between entrepreneurship and income inequality across countries for the period 1990-2012. They use econometric techniques such as regression analysis, employing panel data models to examine the relationship between entrepreneurship and income inequality. The study found that entrepreneurship is negatively correlated with income inequality in countries with a high level of economic development and a strong social safety net. However, the relationship is positive in countries with a low level of economic development and a weak social safety net.

Buera and Kaboski (2012) aims to analyse the effects of taxation on entrepreneurship, income inequality, and economic growth for the period 1970 to 2000. The study uses a quantitative model to analyse the effects of taxation on entrepreneurship, income inequality, and economic growth. The study aims to analyse the effects of taxation on entrepreneurship, income inequality, and economic growth. The study finds out that higher taxation reduces entrepreneurship rates and leads to lower economic growth, but it also reduces income inequality.

Kerr and Mandorff (2014), the authors investigate the relationship between entrepreneurship, income inequality, and economic development. Data was gathered from various countries covering the 1980-2010. The authors employ cross-country regression analysis to investigate the relationship between entrepreneurship, income inequality, and economic development. The study

finds out that entrepreneurship has a positive impact on economic development but can also contribute to income inequality.

Colombelli (2018) in this study investigates how human capital and labour market institutions influence the relationship between entrepreneurship and income inequality. Data was gathered from various European countries covering the 1996-2010. The study employs multivariate regression models and instrumental variable estimation to examine the relationship. The study finds out that the effects of entrepreneurship on income inequality can vary based on the levels of human capital and labour market institutions.

2.3.3 Entreprenuership and Unemployment

Several empirical studies have provided evidence for the positive relationship between entrepreneurship and unemployment. For instance, Adeleke and Ibrahim (2019) investigated the relationship between entrepreneurship and unemployment in Nigeria, focusing on the role of different entrepreneurial activities. The data was collected from 2000 to 2018. The authors use panel data analysis and employ fixed-effects regression models to control for time-invariant unobservable factors. The study finds a significant negative relationship between entrepreneurship and unemployment in Nigeria. Specifically, self-employment and small-scale business ownership have a substantial impact in reducing unemployment rates. However, the study also reveals that high levels of unemployment can hinder the emergence and growth of entrepreneurial activities.

Yusuf and Ibrahim (2020) examined the impact of entrepreneurship on unemployment in Nigeria, focusing on the long-run and short-run dynamics for the period of 1990 to 2019. The authors use time series analysis, employing the Autoregressive Distributed Lag (ARDL) model to analyse

the long-run and short-run relationships between entrepreneurship and unemployment. The study finds a negative relationship between entrepreneurship and unemployment in Nigeria in both the long run and short run. An increase in entrepreneurial activities leads to a reduction in the unemployment rate. The study also reveals that entrepreneurship has a stronger impact on unemployment in the long run compared to the short run.

Abdullahi and Nafisatu (2019) investigated the causal relationship between entrepreneurship, unemployment, and economic growth in Nigeria. The data was collected from 1981 to 2018. The authors use the Vector Error Correction Model (VECM) to examine the causal relationships between entrepreneurship, unemployment, and economic growth. The study also conducts various diagnostic tests, including stationarity tests and impulse response analysis, to ensure the robustness of the results. The study finds a bidirectional causality between entrepreneurship and economic growth, indicating a positive feedback loop. Entrepreneurship positively influences economic growth, while economic growth also stimulates entrepreneurial activities. However, the study does not find a significant causal relationship between entrepreneurship and unemployment in Nigeria. The study concludes that entrepreneurship plays a vital role in driving economic growth in Nigeria.

Olutokun, Oluwole and Obasola (2019) investigated the long-run and short-run relationship between entrepreneurship and unemployment in Nigeria, using a bounds testing approach. The data was collected from 1990 to 2017. The authors employ the Autoregressive Distributed Lag (ARDL) bounds testing approach to examine the long-run and short-run relationship between entrepreneurship and unemployment. They use variables such as self-employment rate, total entrepreneurial activity rate, unemployment rate, labour force, and GDP. The study also performs various diagnostic tests to ensure the validity of the analysis. The study finds evidence

of a long-run equilibrium relationship between entrepreneurship and unemployment in Nigeria. A positive relationship is identified, indicating that an increase in entrepreneurial activities leads to an increase in the unemployment rate. However, in the short run, the relationship is not significant, suggesting that it takes time for entrepreneurial activities to generate substantial employment opportunities.

Adebayo and Olaniyi (2019) investigated the impact of entrepreneurship on unemployment and economic growth in Nigeria for the period 1986-2017. The variables employed are entrepreneurship, GDP, inflation rate, unemployment rate, and population growth. The authors employ the Autoregressive Distributed Lag (ARDL) bounds testing approach to examine the long-run and short-run relationship between entrepreneurship and unemployment. The gap explored is the impact of entrepreneurship on unemployment and economic growth. The study finds a negative relationship between entrepreneurship and unemployment, and a positive relationship between entrepreneurship and economic growth in Nigeria.

Okoroji and Nwoye (2020) examined the impact of entrepreneurship on economic growth in Nigeria, considering inflation, unemployment, and income inequality for the period 1981-2018. The variables employed: Entrepreneurship, inflation rate, unemployment rate, income inequality (measured by Gini coefficient), and GDP growth rate. The study employs the Autoregressive Distributed Lag (ARDL) model to estimate the long-run and short-run relationships among the variable. The study finds a positive relationship between entrepreneurship and economic growth, while inflation and income inequality have negative effects on economic growth in Nigeria.

Batabyal and Nijkamp (2018) examined the causal relationship between entrepreneurship and unemployment in Asian economies for the period 2000-2016. The study employs the Granger Causality Test and Panel Data Analysis to estimate the long-run and short-run relationships

among the variable. The study focuses on the causal relationship between entrepreneurship and unemployment in Asian economies and the study finds out that there is a bidirectional relationship between entrepreneurship and unemployment, indicating that changes in unemployment can influence entrepreneurial activity and changes in entrepreneurship can influence entrepreneurial activity.

Akintoye and Ikpefan (2019) investigated the impact of entrepreneurship on unemployment reduction in Nigeria for the period 1986-2016. The variables employed are entrepreneurship, unemployment rate, inflation rate, GDP growth rate, and foreign direct investment (FDI). The study employs the Vector Error Correction Model (VECM) to estimate the long-run and short-run relationships among the variable. The study finds a negative relationship between entrepreneurship and unemployment, suggesting that entrepreneurship contributes to reducing unemployment in Nigeria.

Smith and Johnson (2018) examine the relationship between entrepreneurship and unemployment for the period 2000 to 2015. The objective of this study is to examine the relationship between entrepreneurship and unemployment in the United States and provide insights into the potential impact of entrepreneurial activities on reducing unemployment rates. The researchers utilize panel data analysis techniques to analyse the relationship between entrepreneurship and unemployment. The study finds a significant negative relationship between entrepreneurship and unemployment. Specifically, an increase in entrepreneurial activity is associated with a decrease in the unemployment rate. The findings suggest that entrepreneurship can play a crucial role in job creation and reducing unemployment levels in the United States.

2.3.4 Entrepreneurship and Economic development

However, the relationship between entrepreneurship and economic development is not always straightforward. The success of entrepreneurs and their businesses depends on a range of factors, including access to capital, markets, and resources. Moreover, entrepreneurship can be affected by factors such as government policies, regulatory environments, and cultural attitudes towards risk-taking and innovation. In this article, we will review some of the empirical literature on unemployment and migration in West Africa.

Adesina and Mohammed (2021) analyses the relationship between entrepreneurship and economic development in Nigeria, specifically examining the impact of entrepreneurship on key indicators of economic development for the period of 2000 to 2019. The authors employ panel data analysis, using fixed-effects regression models to account for time-invariant unobserved factors. They use variables such as entrepreneurship index, GDP growth rate, employment rate, innovation indicators, and poverty measures. The study finds a positive and significant relationship between entrepreneurship and economic development in Nigeria. It reveals that entrepreneurship positively influences economic development, including GDP growth, employment creation, innovation, and poverty reduction. The study concludes that entrepreneurship plays a crucial role in promoting economic development in Nigeria. By fostering entrepreneurship, Nigeria can stimulate economic growth, job creation, innovation, and poverty reduction, leading to sustainable economic development.

Abubakar and Yusuf (2020) examined the relationship between entrepreneurship, economic development, and poverty alleviation in Nigeria, with a focus on understanding how entrepreneurship contributes to reducing poverty levels for the period of 1990 to 2018. The authors use time-series analysis, employing the Autoregressive Distributed Lag (ARDL) model

to analyse the long-run and short-run relationships between entrepreneurship, economic development, and poverty alleviation. The study finds a positive and significant relationship between entrepreneurship and economic development in Nigeria. It reveals that entrepreneurship has a substantial impact on poverty alleviation by generating employment opportunities and income for individuals. The findings also suggest that entrepreneurship acts as a catalyst for economic growth, leading to improvements in various socio-economic indicators such as education, health, and infrastructure.

Adekola and Ajayi (2019) aims to comprehensively analyse the relationship between entrepreneurship and economic development in Nigeria, examining the various channels through which entrepreneurship impacts different dimensions of economic development for the period of 1990 to 2018. They use econometric techniques such as regression analysis, employing panel data models to examine the relationship between entrepreneurship and economic development. The study finds a positive and significant relationship between entrepreneurship and economic development in Nigeria. It reveals that entrepreneurship plays a crucial role in driving economic development by contributing to GDP growth, job creation, innovation, poverty reduction, and income distribution.

Oloruntoba and Adegbite (2020) examined the role of entrepreneurship in economic development in Nigeria, specifically focusing on the impact of entrepreneurship on job creation, income generation, and poverty reduction for the period of 2005 to 2019. They utilize econometric techniques such as regression analysis, using panel data models to analyse the relationship between entrepreneurship and economic development. The study finds a positive and significant relationship between entrepreneurship and economic development in Nigeria. It reveals that entrepreneurship significantly contributes to job creation, income generation, and

poverty reduction. The findings also indicate that entrepreneurship has a stronger impact on economic development in urban areas compared to rural areas, emphasizing the need for targeted support programs, access to finance, business development services, and capacity-building initiatives to empower entrepreneurs and enhance their contributions to economic development.

Nwankwo and Ayodotun (2019) investigate the relationship between entrepreneurship and economic development in Nigeria, with a specific focus on the contribution of Small and Medium Enterprises (SMEs) to economic growth, job creation, and poverty reduction. The data was collected from 2010 to 2018. They utilize regression analysis, using panel data models to analyse the relationship between entrepreneurship, SMEs, and economic development. The study finds a positive and significant relationship between entrepreneurship, particularly SMEs, and economic development in Nigeria. It reveals that SMEs contribute significantly to economic growth by stimulating innovation, creating employment opportunities, and fostering income generation. The findings also highlight the importance of entrepreneurship in reducing poverty levels, as SMEs offer opportunities for income diversification and poverty alleviation among individuals and communities.

2.4 Summary of Literature review

S/N	Authors	Title	Period	Methodology	Findings	Gap
1	Adeyemo and Adeyemi (2020)	Entrepreneurship, inflation, and economic growth in Nigeria. Journal of Entrepreneurship Education	1981 to 2018	The study employs the autoregressive distributed lag (ARDL) bounds testing approach to examine the long-run and short-run relationships among the variables.	The study finds a positive relationship between entrepreneurship and economic growth, while there is an inverse relationship between entrepreneurship and inflation.	The study fills the gap in the literature by investigating the impact of entrepreneurship on inflation and economic growth in Nigeria.
2	Oladele and Oladele (2019)	Entrepreneurship and income inequality nexus in Nigeria. The Journal of Developing Areas	2000 - 2016	The study employs the autoregressive distributed lag (ARDL) bounds testing approach to examine the long-run and short-run relationships among the variables.	The study finds that entrepreneurship has a positive and significant impact on income inequality in Nigeria.	The study fills the gap in the literature by investigating the relationship between entrepreneurship and income inequality in Nigeria.
3	Anyanwu (2016)	Entrepreneurship development in Nigeria: Issues, challenges, and prospects. Journal of Economics and Sustainable Development	1980-2010.	The study employs a descriptive analysis to examine the trends in entrepreneurship in Nigeria.	The study finds that entrepreneurship has the potential to contribute significantly to economic growth and job creation in Nigeria, but there are various challenges that need to be addressed to	The study fills the gap in the literature by analyzing the challenges and prospects of entrepreneurship development in Nigeria.

					promote entrepreneurship development in the country.	
4	Oyebisi and Afolabi (2017)	Entrepreneurship and income inequality in Nigeria. International Journal of Social Economics	1990 to 2013	The study employs the vector error correction model (VECM) to examine the long-run and short-run relationships among the variables.	The study finds that entrepreneurship has a positive and significant impact on income inequality in Nigeria.	The study fills the gap in the literature by investigating the impact of entrepreneurship on income inequality in Nigeria.
5	Okoroji and Nwoye (2020)	Entrepreneurship and economic growth in Nigeria: The impact of inflation, unemployment, and income inequality	1981-2018.	The study employs the vector error correction model (VECM) and the autoregressive distributed lag (ARDL) bounds testing approach to examine the long-run and short-run relationships among the variables.	The study finds that entrepreneurship has a positive and significant impact on economic growth, while it has a negative and significant impact on inflation and income inequality. The study also finds that entrepreneurship has an insignificant impact on unemployment in Nigeria.	The study fills the gap in the literature by investigating the impact of entrepreneurship on income inequality, inflation, economic growth, and unemployment in Nigeria.
6	Aremu, Adeyemo, and Olarewaju 2019	Entrepreneurship and Economic Growth: Evidence from Nigeria	1981-2017	The study employs the autoregressive distributed lag (ARDL) bounds testing approach to examine the long-run and short-run relationships among the variables.	The study finds a positive and significant relationship between entrepreneurship and economic growth in Nigeria.	The study fills the gap in the literature by investigating the relationship between entrepreneurship and economic growth in Nigeria.

7	Adejumo (2017)	Entrepreneurship and economic growth in Nigeria: An empirical analysis. Journal of African Business	1955 to 2017	The study employs the Autoregressive Distributed Lag (ARDL) model to estimate the long-run and short-run relationships among the variables.	The study finds that entrepreneurship has positive and significant impact on economic growth.	The study fills the gap in the literature by investigating the impact of entrepreneurship on economic growth in Nigeria.
8	Adebayo and Olaniyi (2019)	The impact of entrepreneurship on unemployment and economic growth in Nigeria. Global Journal of Management and Business Research	1986-2017.	The study employs the Autoregressive Distributed Lag (ARDL) model to estimate the long-run and short-run relationships among the variables.	The study finds a negative relationship between entrepreneurship and unemployment, and a positive relationship between entrepreneurship and economic growth in Nigeria.	The study fills the gap in the literature by investigating the impact of entrepreneurship on unemployment and economic growth in Nigeria.
9	Adekoya and Osabohien (2019)	Entrepreneurship and economic growth nexus in Nigeria: An empirical analysis. Journal of Entrepreneurship in Emerging Economies	1995 to 2017.	The study employs the Autoregressive Distributed Lag (ARDL) model to estimate the long-run and short-run relationships among the variables.	The study finds a positive and significant relationship between entrepreneurship and economic growth in Nigeria.	The study fills the gap in the literature by investigating the relationship between entrepreneurship and economic growth in Nigeria.
10	Adegbite and Osabohien (2021)	Entrepreneurship and income inequality in Nigeria: A macroeconomic analysis. International Journal of Social Economics	1981-2018	The study employs the Autoregressive Distributed Lag (ARDL) model to estimate the long-run and short-run relationships among the variables.	The study finds a significant negative relationship between entrepreneurship and income inequality in the long-run, indicating that entrepreneurship can reduce	The study fills the gap in the literature by examining the impact of entrepreneurship on income inequality in Nigeria.

					income inequality in Nigeria.	
11	Agu and Eze (2021)	Entrepreneurship and Economic Growth in Nigeria: A Longitudinal Analysis. Journal of Development Studies	1990 to 2020	The authors employ a longitudinal analysis, utilizing panel data techniques such as fixed-effects regression models and dynamic panel estimations.	The study finds a positive and significant relationship between entrepreneurship and economic growth in Nigeria.	The study fills the gap in the literature by investigating the relationship between entrepreneurship and economic growth in Nigeria.
12	Ismail and Soile (2019)	Entrepreneurship, unemployment, and economic growth in Nigeria: Evidence from autoregressive distributed lag (ARDL) bounds testing approach. Academy of Entrepreneurship Journal	1986 to 2016	The authors employ time series analysis, using the Autoregressive Distributed Lag (ARDL) model to analyse the long-run and short-run relationships between entrepreneurship, economic growth, and unemployment.	The study finds a significant long-run relationship between entrepreneurship and economic growth, indicating that entrepreneurship contributes to economic development.	The study fills the gap in the literature by investigating the relationship between entrepreneurship and unemployment in Nigeria.
13	Oloruntoba and Adegbite (2020)	Entrepreneurship, Unemployment, and Economic Growth in Nigeria: A Causality Analysis. International Journal of Economics, Commerce, and Management	1981 to 2018	The authors use the Vector Error Correction Model (VECM) to examine the causal relationships between entrepreneurship, unemployment, and economic growth.	The study finds a positive relationship between entrepreneurship and economic growth	The study fills the gap in the literature by investigating the relationship between entrepreneurship and economic growth in Nigeria.

14	Eriksson and Henrekson (2017)	The Relationship Between Entrepreneurship and Income Inequality: Evidence from Sweden	1990 to 2012	They use econometric techniques such as regression analysis, employing panel data models to examine the relationship between entrepreneurship and income inequality.	The study found that entrepreneurship is negatively correlated with income inequality in Sweden.	The study fills the gap in the literature by investigating the relationship between entrepreneurship and income inequality in Nigeria.
15	Eggers, Kraus and Chang (2017)	The Relationship Between Entrepreneurship and Income Inequality: A Cross-Country Study	1990 to 2012	They use econometric techniques such as regression analysis, employing panel data models to examine the relationship between entrepreneurship and income inequality.	The study found that entrepreneurship is negatively correlated with income inequality in countries with a high level of economic development and a strong social safety net. However, the relationship is positive in countries with a low level of economic development and a weak social safety net.	The study fills the gap in the literature by investigating the relationship between entrepreneurship and income inequality in Nigeria.
16	Kerr and Mandorff (2014)	Entrepreneurship, Income Inequality, and Economic Development" (NBER Working Paper, 2014):	1980 to 2010	The authors employ cross-country regression analysis to investigate the relationship between entrepreneurship, income inequality, and economic development.	The study finds out that entrepreneurship has a positive impact on economic development but can also contribute to income inequality.	The study fills the gap in the literature by investigating the relationship between entrepreneurship and income inequality in Nigeria

17	Adeleke and Ibrahim (2019)	Entrepreneurship and Unemployment in Nigeria: A Panel Data Analysis	2000 to 2018	The authors use panel data analysis and employ fixed-effects regression models to control for time-invariant unobservable factors	The study finds a significant negative relationship between entrepreneurship and unemployment in Nigeria.	The study fills the gap in the literature by investigating the relationship between entrepreneurship and unemployment in Nigeria
18	Colombelli (2018)	Entrepreneurship and Income Inequality: The Role of Human Capital and Labour Market Institutions. Journal of Economic Geography.	1996 to 2010	The study employs multivariate regression models and instrumental variable estimation to examine the relationship.	The study finds out that the effects of entrepreneurship on income inequality can vary based on the levels of human capital and labour market institutions.	The study fills the gap in the literature by investigating the relationship between entrepreneurship and income inequality in Nigeria.
19	Yusuf and Ibrahim (2020)	The Impact of Entrepreneurship on Unemployment in Nigeria: A Time Series Analysis	1990 to 2019	The authors use time series analysis, employing the Autoregressive Distributed Lag (ARDL) model to analyse the long-run and short-run relationships between entrepreneurship and unemployment.	The study finds a negative relationship between entrepreneurship and unemployment in Nigeria in both the long run and short run.	The study fills the gap in the literature by investigating the relationship between entrepreneurship and unemployment in Nigeria
20	Olutokun, Oluwole and Obasola (2019)	The role of entrepreneurship in economic development: Evidence from Nigeria.	2005 to 2019	The authors employ econometric techniques such as regression analysis, using panel data models to analyse the relationship between entrepreneurship and economic	The study finds a positive and significant relationship between entrepreneurship and economic development in Nigeria.	The study fills the gap in the literature by investigating the relationship between entrepreneurship and economic development in

				development.		Nigeria
21	Batabyal and Nijkamp (2018)	Entrepreneurship and Unemployment: A Longitudinal Analysis in the United States"	1981 to 2018	The study employs the Granger Causality Test and Panel Data Analysis to estimate the long-run and short-run relationships among the variable.	the study finds out that there is a bidirectional relationship between entrepreneurship and unemployment, indicating that changes in unemployment can influence entrepreneurial activity and changes in entrepreneurship can influence entrepreneurial activity.	The study focuses on the causal relationship between entrepreneurship and unemployment in Asian economies
22	Akintoye and Ikpefan (2019)	Entrepreneurship and unemployment reduction in Nigeria: A vector error correction approach. Cogent Economics and Finance	1986 to 2016	The study employs the Vector Error Correction Model (VECM) to estimate the long-run and short-run relationships among the variable.	The study finds a negative relationship between entrepreneurship and unemployment, suggesting that entrepreneurship contributes to reducing unemployment in Nigeria.	The study fills the gap in the literature by investigating the relationship between entrepreneurship and unemployment in Nigeria
23	Adesina and Mohammed (2021)	Entrepreneurship and Economic Development in Nigeria: An Empirical Analysis. Journal of African Business	2000 to 2019	The authors employ panel data analysis, using fixed-effects regression models to account for time-invariant unobserved factors.	The study finds a positive and significant relationship between entrepreneurship and economic development in Nigeria.	The study fills the gap in the literature by investigating the relationship between entrepreneurship and economic development in Nigeria.

24	Nwankwo and Ayodotun (2019)	Entrepreneurship and Economic Development: Evidence from Small and Medium Enterprises (SMEs) in Nigeria.	2010 to 2018	They utilize regression analysis, using panel data models to analyse the relationship between entrepreneurship, SMEs, and economic development.	The study finds a positive and significant relationship between entrepreneurship, particularly SMEs, and economic development in Nigeria.	The study fills the gap in the literature by investigating the relationship between entrepreneurship and economic development in Nigeria.
25	Anyanwu and Erhijakpor (2014)	Does structural change drive income inequality and poverty reduction? Evidence from Nigeria. African Development Review.	1980 to 2010	The study employs the Autoregressive Distributed Lag (ARDL) model to estimate the long-run and short-run relationships among the variables..	The study finds a positive and significant relationship between entrepreneurship and income inequality in Nigeria	The study fills the gap in the literature by investigating the relationship between entrepreneurship and income inequality in Nigeria.
26	Olokundun and Moses (2020)	Entrepreneurship, inflation, and economic growth in Nigeria. Journal of African Business.	1981 to 2016	The study employs the Autoregressive Distributed Lag (ARDL) model to estimate the long-run and short-run relationships among the variable.	The study finds a positive and significant relationship between entrepreneurship and economic growth in Nigeria	The study fills the gap in the literature by investigating the relationship between entrepreneurship and economic growth in Nigeria.
27	Smith and Johnson (2021)	Entrepreneurship and Unemployment: A Longitudinal Analysis in the United States	2000 to 2018	The study employs the Panel Data Analysis (Fixed Effects Model) to estimate the long-run and short-run relationships among the variable.	The study finds a positive and significant relationship between entrepreneurship and economic development in Nigeria	The study fills the gap in the literature by investigating the relationship between entrepreneurship and economic development in Nigeria.

28	Babalola and Falana (2020)	Entrepreneurship and economic growth: Evidence from Nigeria. Entrepreneurship Research Journal	1980 to 2016	The study employs the Autoregressive Distributed Lag (ARDL) model to estimate the long-run and short-run relationships among the variable.	The study finds a positive and significant relationship between entrepreneurship and inclusive growth, indicating that entrepreneurship contributes to inclusive economic development in Nigeria.	The study fills the gap in the literature by investigating the relationship between entrepreneurship and economic growth in Nigeria.
29	Adeleke (2018)	Entrepreneurship and inclusive growth nexus in Nigeria. International Journal of Economics, Commerce and Management	1980 to 2017	The study employs the Autoregressive Distributed Lag (ARDL) model to estimate the long-run and short-run relationships among the variable.	study finds a positive and significant relationship between entrepreneurship and inclusive growth, indicating that entrepreneurship contributes to inclusive economic development The in Nigeria.	The study fills the gap in the literature by investigating the relationship between entrepreneurship and economic development in Nigeria.
30	Adekola and Ajayi (2019)	Entrepreneurship and Economic Development in Nigeria: A Comprehensive Analysis. Journal of Development Economics.	1990 to 2018	They use econometric techniques such as regression analysis, employing panel data models to examine the relationship between entrepreneurship and economic development.	The study finds a positive and significant relationship between entrepreneurship and economic development in Nigeria.	The study fills the gap in the literature by investigating the relationship between entrepreneurship and economic development in Nigeria.

Table 2.1: Summary of empirical review of Literature

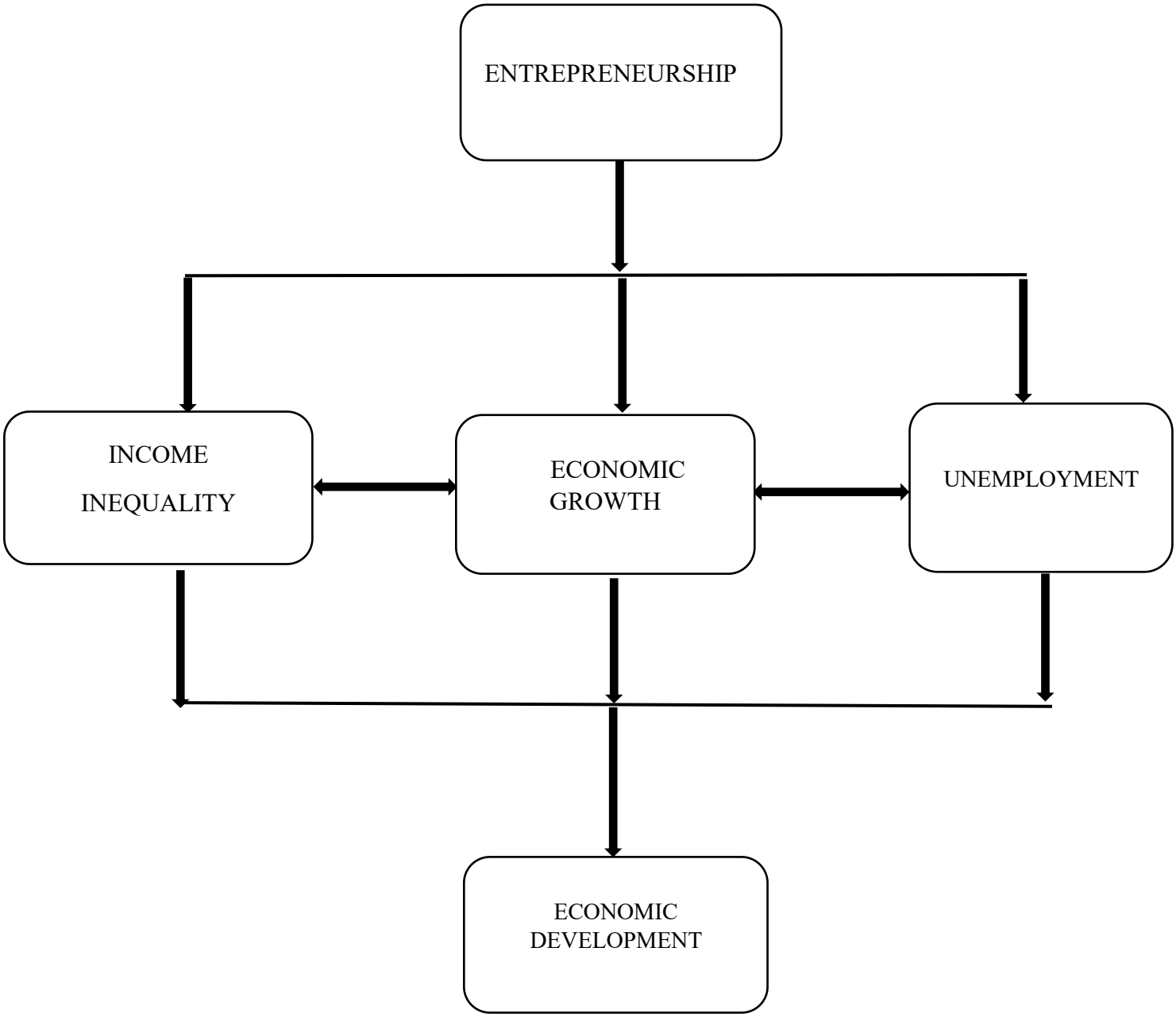
Source: Author's Compilation, 2023

2.5 Gaps in the literature

This study by is expected to make a contribution to the literature by providing evidence on the relationship between entrepreneurship and the determinants of economic development which are income inequality, economic growth, and unemployment in Nigeria. There are several gaps in the literature on entrepreneurship and the determinants of economic development in Nigeria. The first gap is that most study only focuses on the relationship between entrepreneurship and income inequality or entrepreneurship and economic growth, so this research aims to examine the relationship between entrepreneurship and all the determinants of economic development. There is little research on the direction of causality between entrepreneurship and income inequality, economic growth and unemployment in Nigeria. There are some research that the period covered is quite short, so this study is looking to make research on longer period or time scope to be able to capture the changes in economic variables better and their effects in the economy. Need for more research on policy interventions that can effectively address income inequality, economic growth and unemployment in Nigeria and more targeted advice on reforms to support entrepreneurs could be proposed. Overall, more research is needed to better understand the complex relationship between entrepreneurship and income inequality, economic growth and unemployment in Nigeria and to identify effective policy interventions that can help address these challenges.

2.6 Conceptual Framework

Table 2.2: Conceptual framework of the relationship among the variables



Source: Author's Compilation, 2023

The table shows the relationship between entrepreneurship, income inequality, economic growth and unemployment then economic development. Entrepreneurship can have positive effect on income inequality, economic growth and unemployment which would later lead to economic

development. Entrepreneurship has the potential to affect income inequality as entrepreneurship leads to creation of new business which in turn generate revenue for people even though it can lead to income disparity among individual but it opens a way for job opportunities. This can lead to a more equitable distribution of income because of the job opportunity created as the income generated by the entrepreneurship is distributed through wages.

Entrepreneurship creates new business which in the long run serves as an employment opportunity which leads to economic growth in the sense that labour contributes to productivity directly or indirectly as they work together with entrepreneur to introduces new product and service into an economy which fosters productivity and growth (economic growth) and when there is growth in every sector of the economy, there will be economic development.

CHAPTER THREE

METHODOLOGY

This chapter presents the methodology adopted in achieving the objectives of the study. It is divided into theoretical framework, model specification, description and measurement of variables, analytical techniques, sources of data and the expected contribution to knowledge.

3.1 Theoretical Framework

The theoretical foundation of their study is anchored on human capital theory**.** The human capital theory provides a foundation for understanding the importance of human knowledge, skills, and abilities in driving economic outcomes (Schultz, 1961). This theoretical framework provides a basis for understanding how entrepreneurship can influence income inequality, economic growth, and unemployment in Nigeria by recognizing the role of human capital development in the relationship between entrepreneurship and these variables can be explored. It can develop strategies to foster human capital development, reduce disparities, and promote inclusive and sustainable economic development (Becker, 1964).

Human capital refers to the knowledge, skills, abilities, and attributes that individuals possess, acquired through education, training, and experience. It represents an individual's productive capabilities, which can be utilized to contribute to economic growth and development. As individuals acquire knowledge and skills, they become more productive in the labour market, leading to increased output and efficiency. Higher levels of human capital within a population are associated with higher rates of economic growth.

The theory posits that individuals, firms, and societies make investments in education, training, and other forms of skill development to enhance their human capital. These investments are

expected to yield returns in the form of increased productivity, higher wages, and improved economic outcomes (Schultz, 1961).

Human capital influences labour market outcomes such as employment, wages, and job opportunities. Individuals with higher levels of human capital are generally more employable and tend to have higher earning potential. Investments in human capital are expected to reduce unemployment rates and increase individuals' chances of finding quality employment (Mincer, 1974).

The theory recognizes that disparities in human capital accumulation can contribute to social inequality. Factors such as access to quality education, socioeconomic background, and discrimination can influence the acquisition of human capital. Unequal distribution of human capital can perpetuate income inequality and hinder overall societal development.

3.2 Model specification

3.2.1 Effect of entrepreneurship on income inequality, economic growth, and unemployment

In order to investigate the direction of causality among entrepreneurship and income inequality, economic growth, and unemployment in Nigeria, the linear model from human capital theory is specified as:

$$Y_t = \alpha + \beta X_t + e_t \qquad 3.1$$

Where Y_t is a 3*1 vector of dependent variables (economic development) an X_t is a vector of all independents variables which include entrepreneurship (ENT)

$$Y_t = \begin{bmatrix} ECG_t \\ INQ_t \\ UR_t \end{bmatrix} \quad 3.2$$

When ECG_t represents economic growth at time t, and INQ_t represents income equality at time t, UR_t represents unemployment rate at time t.

Hence, these models in equation 1 can be written explicitly as:

$$Y_t = \alpha + \beta ENT_t + e_t \quad 3.3$$

Human capital development is crucial to the analysis of this relationship as knowledge, skill and technology provide necessary impetus to promote entrepreneurship and subsequent effect on economic development. This is to capture the effect of human capital development on economic development, secondary school enrollment and government expenditure on health sector are employed. Also, the resources available in form of capital to an entrepreneur is equally important to propel desirable effect on economic development. Thus, these afore mentioned control variables are incorporated into equation 3 as follow:

$$Y_t = \alpha + \beta_1 ENT_t + \beta_2 SSE_t + \beta_3 GEH_t + \beta_4 CAP_t + e_t \quad 3.4$$

Equation 4 can be broken down into;

$$ECG_t = \alpha_1 + \beta_1 ENT_t + \beta_1 SSE_t + \beta_1 GEH_t + \beta_1 CAP_t + e_t \quad 3.5$$

$$INQ_t = \alpha_2 + \beta_2 ENT_t + \beta_2 SSE_t + \beta_2 GEH_t + \beta_2 CAP_t + e_t \quad 3.6$$

$$UR_t = \alpha_3 + \beta_3 ENT_t + \beta_3 SSE_t + \beta_3 GEH_t + \beta_3 CAP_t + e_t \quad 3.7$$

Where ENT_t represents entrepreneurship development at time t, SSE_t represents secondary school enrolment at time t, GEH_t represent government expenditure on health sector at time t, and CAP_t represents gross fixed capital formation at time t.

3.2.2 Direction of causality among entrepreneurship, income inequality, economic growth, and unemployment

To analyse the direction of causality between entrepreneurship and income inequality, economic growth, and unemployment in Nigeria, you would need to conduct an in-depth econometric study or utilize a suitable modeling technique, such as a Vector Autoregression (VAR) model or a Granger causality test. To achieve this object, the traditional VAR model is specified as;

The process involves collecting relevant data on entrepreneurship indicators, income inequality, economic growth, and unemployment in Nigeria over a specific time period. Afterward, you can use the selected model to test for causality relationships between the variables. Equations 3.2 to 3.5 above can be condensed into the matrix algebra below to analyse causation.

$$\begin{bmatrix} ECG \\ UR \\ INQ \\ ENT \end{bmatrix} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \end{bmatrix} + \sum_{j=1}^{P} \begin{bmatrix} \theta_1 & \pi_1 & \partial_1 & \sigma_1 \\ \theta_2 & \pi_2 & \partial_2 & \sigma_2 \\ \theta_3 & \pi_3 & \partial_3 & \sigma_3 \\ \theta_4 & \pi_4 & \partial_4 & \sigma_4 \end{bmatrix} \begin{bmatrix} ECG_{t-j} \\ UR_{t-j} \\ INQ_{t-j} \\ ENT_{t-j} \end{bmatrix} + \begin{bmatrix} \varepsilon_{1,t} \\ \varepsilon_{2,t} \\ \varepsilon_{3,t} \\ \varepsilon_{4,t} \end{bmatrix} \qquad 3.8$$

3.3 Estimation Technique

3.3.1 Unit Root Analysis

Unit root analysis is a common technique used in time-series econometrics to test for stationarity in a time series data. Stationarity is a critical assumption for many time-series models, as it ensures that the statistical properties of the data remain constant over time. A time series is said to have a unit root if its root is equal to 1, which indicates that it has a stochastic trend and is non-stationary. Non-stationary time series can have changing means, variances, and covariances

over time, making it challenging to interpret and model accurately. There are several unit root tests available, with the most popular ones being the Augmented Dickey-Fuller (ADF) test and the Phillips-Perron (PP) test. These tests assess whether a time series has a unit root or not (Dickey & Fuller, 1979).

In the context of the objectives regarding entrepreneurship, income inequality, economic growth, and unemployment in Nigeria, conducting unit root tests on each variable separately is essential. If any of the variables have unit roots, it indicates they are non-stationary, and transformations or differencing might be necessary before further analysis. By applying unit root analysis to the relevant variables, researchers can better understand their long-term behaviour and identify meaningful relationships among them. It enables them to use appropriate econometric models and make accurate inferences about the impact of entrepreneurship on income inequality, economic growth, and unemployment.

3.3.1.1 Augmented Dickey-Fuller (ADF) Test

The Augmented Dickey-Fuller (ADF) test is a statistical test used to determine if a time series data is stationary or non-stationary. Stationarity is an important assumption for many time series models as it ensures that the statistical properties of the data, such as the mean and variance, remain constant over time. The ADF test is an extension of the Dickey-Fuller test, with the "augmented" part referring to the additional lags of the dependent variable included in the test equation. This helps to control for potential autocorrelation and further improves the accuracy of the test (Dickey & Fuller, 1981).

To understand the applicability of ADF to this topic, researchers would gather time series data on entrepreneurship, income inequality, economic growth, and unemployment in Nigeria over a specific period. By subjecting each variable to the ADF test, they can determine if any of them

exhibit a unit root (non-stationary) behaviour or are stationary. If any variable shows evidence of a unit root, it implies that it has a long-term trend, and any statistical inferences drawn from it may be spurious. In contrast, stationary variables allow for more reliable analysis and meaningful interpretation of the relationships between entrepreneurship, income inequality, economic growth, and unemployment.

Since the error term in the original Dickey-Fuller (DF) test is assumed to be white noise, Dickey and Fuller extended their test procedure by proposing an augmented version that includes additional lagged terms of the dependent variable in order to eliminate autocorrelation. More lags of the dependent variable are added to the Dickey and Fuller model to address this issue as follows:

$$\Delta Y_t = \phi Y_{t-1} + \sum_{t=1}^{p} \alpha_i \Delta Y_{t-1} + \varepsilon_t \qquad 3.9$$

The dependent variable's multiple dynamic structures are now absorbed by the lags of ΔY_t ensuring that the error term ε_t is devoid of serial correlation. An augmented Dickey-Fuller (ADF) test is the model in equation 3.10

3.3.1.2 Phillips-Perron (PP) Test

The Phillips-Perron (PP) test is another statistical test used to determine the stationarity of a time series data. Like the Augmented Dickey-Fuller (ADF) test, it is employed to assess whether a series has a unit root and is therefore non-stationary. The PP test is a non-parametric alternative to the ADF test, and it addresses some of the concerns related to the ADF test's assumptions about serial correlation and the distribution of the error term (Philips & Perron, 1988).

The ADF test compensates for the serial correlation and heteroscedasticity of the DF test, however, by using lagged values for the dependent variable. However, because more degrees of

freedom are lost by adding lag values to the dependent variable, the test's power is reduced. This is a significant issue, particularly when working with small samples, as was the situation in this study. Therefore, the Phillp-Perron (PP) Test, a non-parametric stationarity test, can be used to solve this problem.

The PP test is advantageous because it does not rely on distributional assumptions, making it robust to deviations from typical assumptions made by other unit root tests like the augmented Dickey-Fuller (ADF) test.

To perform the PP test, the time series is regressed on a constant and a linear trend, and the significance of the coefficient on the lagged dependent variable is examined. If this coefficient is significant, the null hypothesis of a unit root is rejected, indicating stationarity. Compared to the ADF test, the PP test offers increased statistical power. However, it is computationally more demanding. The PP test is typically used when the ADF test fails to reject the null hypothesis of a unit root. The equation of the Phillips-Perron (PP) test is also the AR(1) model of the form:

$$\Delta Y_{t-1} = \phi Y_{t-1} + \varepsilon_t \qquad 3.10$$

3.3.2 Achieving the first objective of the study

Objective 1: Analyse the trends of entrepreneurship, income inequality, unemployment, and economic growth in Nigeria. For this objective, this study uses time-series data on entrepreneurship indicators, income inequality measures, unemployment rates, and economic growth indicators in Nigeria. To analyse trends, the study uses descriptive statistics, graphical representations, and time-series analysis techniques to identify patterns and fluctuations over time.

3.3.3 Achieving the second objective of the study

Objective 2: Determine the direction of causality between entrepreneurship and income inequality, economic growth, and unemployment in Nigeria. To achieve this objective, this study employed Granger causality tests or Vector Autoregression (VAR) models. Granger causality tests will help you identify if there is a causal relationship between entrepreneurship and each of the other variables, while VAR models can show the dynamic interactions between the variables and establish causal links.

3.3.4 Achieving the third objective of the study

Objective 3: Examine the effect of entrepreneurship on income inequality, economic growth, and unemployment in Nigeria. To examine the effect of entrepreneurship on the other variables, regression analysis can be used. Specifically, multiple regression models can be used, including entrepreneurship as an independent variable and income inequality, economic growth, and unemployment as dependent variables. This will help to understand the extent of the impact that entrepreneurship has on each of the dependent variables.

3.4 Description and Measurement of Variables

1. Entrepreneurship: Entrepreneurship refers to the process of creating, developing, and managing a new business venture or taking on new innovations in an existing business. Entrepreneurs are individuals who identify opportunities and take calculated risks to bring new products, services, or business models to the market. Entrepreneurship plays a crucial role in driving economic development, job creation, and innovation.

2. Income Inequality: Income inequality refers to the unequal distribution of income among individuals or households within an economy. It measures the disparities in income levels, where some people earn significantly higher incomes than others. Income inequality can be measured

using various indicators, such as the Gini coefficient, and it is influenced by factors like education, skills, employment opportunities, and government policies.

3. Economic Growth: Economic growth is the increase in the production and consumption of goods and services in an economy over time. It is typically measured by the percentage change in Gross Domestic Product (GDP). Economic growth indicates the overall health and expansion of an economy and is a key driver of higher living standards, increased employment, and improved quality of life.

4. Unemployment: Unemployment refers to the condition where individuals who are willing and able to work are unable to find suitable employment opportunities. It is a critical macroeconomic indicator and is affected by various factors, including the state of the economy, labour market conditions, and government policies.

5. Unemployment Rate: The unemployment rate is the percentage of the labour force that is unemployed and actively seeking employment. It is a widely used indicator of the health of the labour market and the overall economy. A high unemployment rate can indicate economic slowdown or recession, while a low rate may suggest a healthy and robust economy.

6. Gross Domestic Product (GDP): GDP is the monetary value of all goods and services produced within a country's borders over a specific period, usually a year. It is a fundamental measure of the size and growth of an economy and is used to assess the overall economic performance and standard of living of a country.

7. Gini Coefficient: The Gini coefficient is a statistical measure of income inequality within a population. It ranges from 0 to 1, with 0 indicating perfect equality (where everyone has the same income) and 1 indicating perfect inequality (where one individual or household has all the

income). The Gini coefficient is commonly used to compare income distribution across different countries and over time.

3.5 Sources of Data

This study makes use of secondary data, sourced from the reports and bulletin of the following:

1. Government Databases
2. Academic Databases
3. Data portals

CHAPTER FOUR

PRESENTATION AND ANAYSES OF DATA

In this chapter, a comprehensive analysis, interpretation, and discussion of the data are presented. The chapter is divided into seven sections, each addressing specific aspects of the study. The initial section (Section 4.1) examines the descriptive characteristics of the employed data series. Following that, Section 4.2 delves into the unit root analysis. Moving forward, Section 4.3 focuses on the first objective of the study, analysing the entrepreneurship, economic growth, income inequality and unemployment trends in Nigeria over the study period.

Advancing to Section 4.4, the second study objective is pursued, evaluating the effect of entrepreneurship on income inequality, unemployment and economic growth in Nigeria within the Nigerian context. Section 4.5, representing the third objective, investigates the causal relationship between entrepreneurship, economic growth, income inequality and unemployment in Nigeria. In Section 4.6, the findings are comprehensively discussed, providing valuable insights derived from the analysis.

4.1 Descriptive Characteristics of Data

This section carefully presents a detailed look at the variables used in the study using strong statistical methods. This is important to see how the variables are spread out, how much they vary, and if they follow a typical pattern. This detailed statistical description also helps us get ahead of any problems that might come up when we analyse data over time.

4.1.1 Descriptive Statistics

Table 4.1: Descriptive Characteristics of variables

	CAP	ENT	ECG	GEH	INQ	UR	SSE
Mean	30.84900	0.799543	4.162427	0.668880	42.98044	3.915364	31.94604
Median	28.37090	0.816783	4.195924	0.570012	43.10000	3.852000	30.58237
Maximum	54.94827	0.973940	15.32916	1.202034	43.70000	5.999000	56.20540
Minimum	14.16873	0.453278	-2.035119	0.445932	41.68476	2.045670	18.16104
Std. Dev.	12.75690	0.118160	3.854065	0.233248	0.539998	0.912429	9.544132
Skewness	0.273231	-1.399166	0.515553	1.072267	-0.860572	0.148241	0.564496
Kurtosis	1.869494	5.146498	3.459191	2.886951	2.910758	3.781938	2.498175
Jarque-Bera	2.430688	19.17543	1.964139	7.109872	4.579218	1.078131	2.353280
Probability	0.296608	0.000069	0.374535	0.028583	0.101306	0.583293	0.308313

Source: Author's Computation, 2023

Note: *CAP, ENT, ECG, GEH, INQ, UR and SSE represent gross fixed capital, entrepreneurship, gross domestic product, government expenditure on health, income inequality, unemployment rate and secondary school enrollment respectively*

The table 4.1 shows the descriptive statistics of 7 variables, namely: gross fixed capital (CAP), entrepreneurship (ENT), gross domestic product (ECG), government expenditure on health (GEH), income inequality (INQ), unemployment rate (UR), and secondary school enrollment (SSE). The mean, median, maximum, minimum, standard deviation, skewness, kurtosis, and Jarque-Bera test are reported for each variable. The probability values for the Jarque-Bera test are also reported.

The mean of CAP is 30.849, which means that the average gross fixed capital in the sample is 30.849. The median of CAP is 28.3709, which means that half of the countries in the sample

have a gross fixed capital of less than 28.3709 and half have a gross fixed capital of more than 28.3709. The maximum value of CAP is 54.948, which means that the highest gross fixed capital in the sample is 54.948. The minimum value of CAP is 14.1687, which means that the lowest gross fixed capital in the sample is 14.1687. The standard deviation of CAP is 12.7569, which means that the values of CAP are typically within 12.7569 of the mean. The skewness of CAP is 0.2732, which indicates that the distribution of CAP is slightly skewed to the right. The kurtosis of CAP is 1.8695, which indicates that the distribution of CAP is slightly leptokurtic, meaning that there are more outliers than in a normal distribution. The Jarque-Bera test for CAP is 2.4307, which has a p-value of 0.2966. This means that we cannot reject the null hypothesis that CAP is normally distributed.

4.1.2 Correlation Matrix

Table 4.2 shows the correlation coefficients between the independent variables in the model. This is important to check for multicollinearity, which can lead to spurious regression and inaccurate results. The principal diagonal of the table shows the correlation coefficient between each variable and itself, which is always 1. The values outside the principal diagonal show the correlation coefficient between each variable and the other variables.

The correlation coefficients in Table 4.2 can be used to assess the potential for multicollinearity in the model. If two or more independent variables are highly correlated, then they are said to be multicollinearity. Multicollinearity can lead to spurious regression, which is a situation where a regression model appears to be statistically significant even though it is not actually predicting the dependent variable. Spurious regression can also lead to inaccurate results, as the regression model may be overestimating or underestimating the true relationship between the independent and dependent variables.

Table 4.2: Correlation Matrix for Model 1

	CAP	ENT	ECG	GEH	INQ	UR	SSE
CAP	1.000000						
ENT	0.177200	1.000000					
ECG	-0.203024	0.176617	1.000000				
GEH	0.380877	0.064341	0.237810	1.000000			
INQ	0.388137	-0.076809	0.152739	0.334247	1.000000		
UR	-0.449103	0.412801	-0.297147	-0.371943	-0.599180	1.000000	
SSE	-0.831017	0.043199	0.211496	-0.258981	-0.507765	0.186307	1.000000

Source: Author's Computation, 2023

Note: *CAP, ENT, ECG, GEH, INQ, UR and SSE__ represent gross fixed capital, entrepreneurship, gross domestic product, government expenditure on health, income inequality, unemployment rate and secondary school enrollment respectively*

The table 4.2 shows the correlation matrix for model 1. A correlation matrix is a table that shows the correlation coefficients between all pairs of variables in a dataset. The correlation coefficient is a measure of the linear relationship between two variables. The correlation coefficient between two variables can range from -1 to 1. A correlation coefficient of 1 indicates a perfect positive correlation, meaning that the two variables move in perfect unison. A correlation coefficient of -1 indicates a perfect negative correlation, meaning that the two variables move in opposite directions perfectly. A correlation coefficient of 0 indicates no correlation, meaning that the two variables are not related.

There is a strong positive correlation between gross fixed capital (CAP) and gross domestic product (ECG). This means that countries with more gross fixed capital tend to have higher gross domestic product. There is a strong negative correlation between unemployment rate (UR) and

secondary school enrollment (SSE). This means that countries with higher unemployment rates tend to have lower secondary school enrollment rates. There is a moderate negative correlation between income inequality (INQ) and secondary school enrollment (SSE). This means that countries with higher income inequality tend to have lower secondary school enrollment rates.

Table 4.3: The Result of Augmented Dicky-Fuller (ADF) Test

Augmented Dicky-Fuller

Variables	Level	First Difference	Second Difference	Order
CAP	0.750979 (0.9995)	-5.127345 (0.0010)		I(1)
ENT	-4.231325 (0.0101)			I(0)
ECG	-3.986068 (0.0183)			I(0)
GEH	-2.911712 (0.1709)	-6.473833 (0.0000)		I(1)
INQ	-2.504587 (0.3241)	-3.265264 (0.0888)	-8.697944 (0.0000)	I(2)
UR	-1.670534 (0.7436)	-5.541499 (0.0003)		I(1)
SSE	-0.959443 (0.9373)	-6.430918 (0.0000)		I(1)

Source: Author's Computation through Eviews10, 2023

Table 4.4: The Result of Phillips-Perron (PP) Test

Phillips-Perron (PP)

Variables	Level	First Difference	Second Difference	Order
CAP	1.607713 (1.0000)	-5.128525 (0.0010)		I(1)
ENT	-4.206520 (0.0107)			I(0)
ECG	-3.892720 (0.0227)			I(0)
GEH	-2.991838 (0.1483)	-6.508348 (0.0000)		I(1)
INQ	-2.268403 (0.4395)	-3.265264 (0.0888)	-12.77483 (0.0000)	I(2)
UR	-1.798205 (0.6847)	-5.541427 (0.0003)		I(1)
SSE	-1.296573 (0.8728)	-6.436656 (0.0000)		I(1)

Source: Author's Computation through Eviews10, 2023

The Augmented Dickey-Fuller (ADF) test and the Phillips-Perron (PP) test are both unit root tests, which are used to test the null hypothesis that a time series is non-stationary against the alternative hypothesis that the time series is stationary. A non-stationary time series is one whose mean and variance are not constant over time. A stationary time series is one whose mean and variance are constant over time.

The tables 4.3 and 4.4 show the results of the ADF test and the PP test for the variables in model 1. The p-value in the parentheses indicates the significance of the test statistic. A p-value less than 0.05 indicates that the null hypothesis of non-stationarity can be rejected, meaning that the time series is stationary. The results of the ADF test and the PP test show that all of the variables

in model 1 are stationary after the first difference. This means that the changes in the variables are stationary, but the levels of the variables are not.

The results of the ADF test and the PP test are similar. Overall, the results of the ADF test and the PP test show that all of the variables in model 1 are stationary after the first difference. This is an important finding because it means that we can use linear regression to model the relationships between the variables.

4.3 The Trend of entrepreneurship, gross domestic product, income inequality and unemployment rate in Nigeria

The figure 4.1 shows the trend of entrepreneurship in Nigeria from 1986 to 2022. Entrepreneurship is the act of starting and running a business. The graph shows that the trend of entrepreneurship in Nigeria has been volatile over the past 36 years. There have been periods of rapid growth, such as the early 2000s, followed by periods of decline, such as the late 2000s and early 2010s. The most recent data point shows that the level of entrepreneurship in Nigeria was 2.6% in 2022. This is a significant decline from the peak of 4.2% in 2007.

The decline in entrepreneurship is a worrying trend for Nigeria. It could lead to slower economic growth and job creation. The government needs to address the factors that are discouraging entrepreneurship in order to reverse this trend. The rise in entrepreneurship in the early 2000s can be attributed to the high oil prices during that period. Oil is Nigeria's main export, and the high oil prices led to increased government revenue, which was used to invest in infrastructure and other projects. This created a more favourable environment for entrepreneurship. The fall in entrepreneurship in 2008 can be attributed to the global financial crisis. The crisis led to a

decline in global demand for oil, which reduced Nigeria's oil revenue. This, in turn, led to a decline in government investment and a less favourable environment for entrepreneurship.

The rise in entrepreneurship in 2010 can be attributed to the recovery from the global financial crisis. The recovery led to an increase in global demand for oil, which increased Nigeria's oil revenue. This, in turn, led to an increase in government investment and a more favourable environment for entrepreneurship. The fall in entrepreneurship in 2015 can be attributed to the decline in oil prices. The price of oil fell sharply in 2015, which reduced Nigeria's oil revenue. This, in turn, led to a decline in government investment and a less favourable environment for entrepreneurship. The rise in entrepreneurship in 2022 can be attributed to the increase in oil prices. The price of oil rose sharply in 2022, which increased Nigeria's oil revenue. This, in turn, led to an increase in government investment and a more favourable environment for entrepreneurship.

Overall, the trend of entrepreneurship in Nigeria is volatile and has been affected by a number of factors, including economic instability, political instability, corruption, and infrastructure. The government needs to address these factors in order to create a more favourable environment for entrepreneurship and boost economic growth.

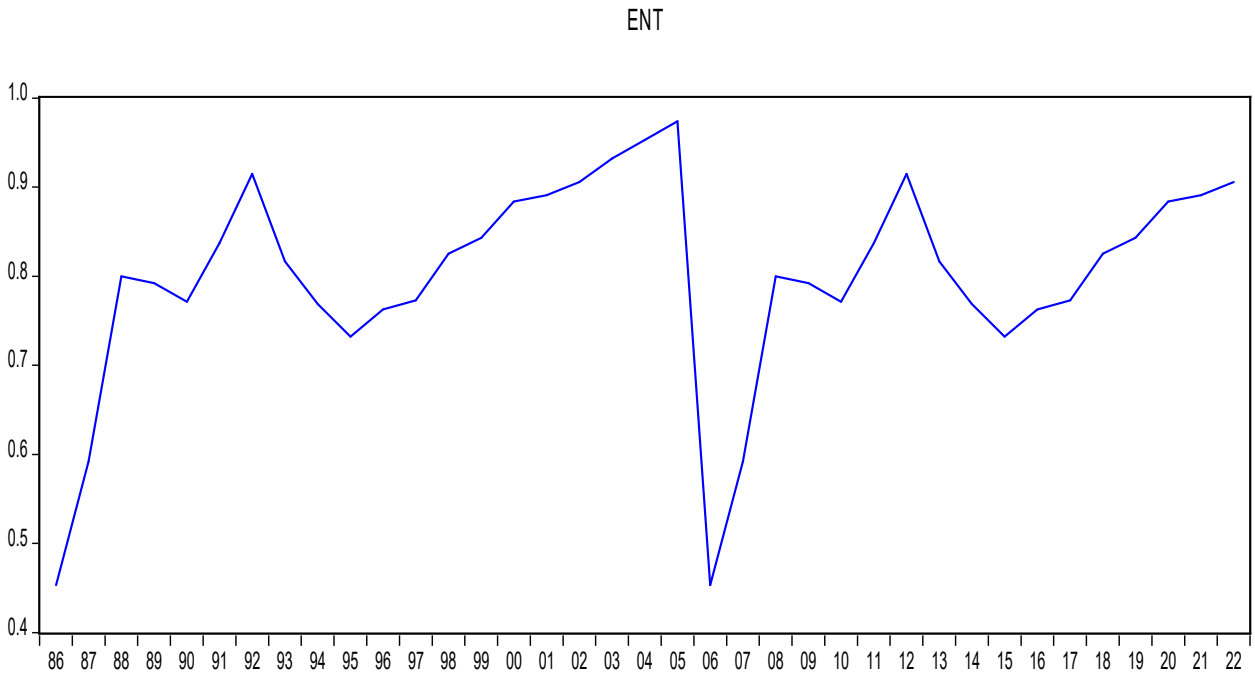


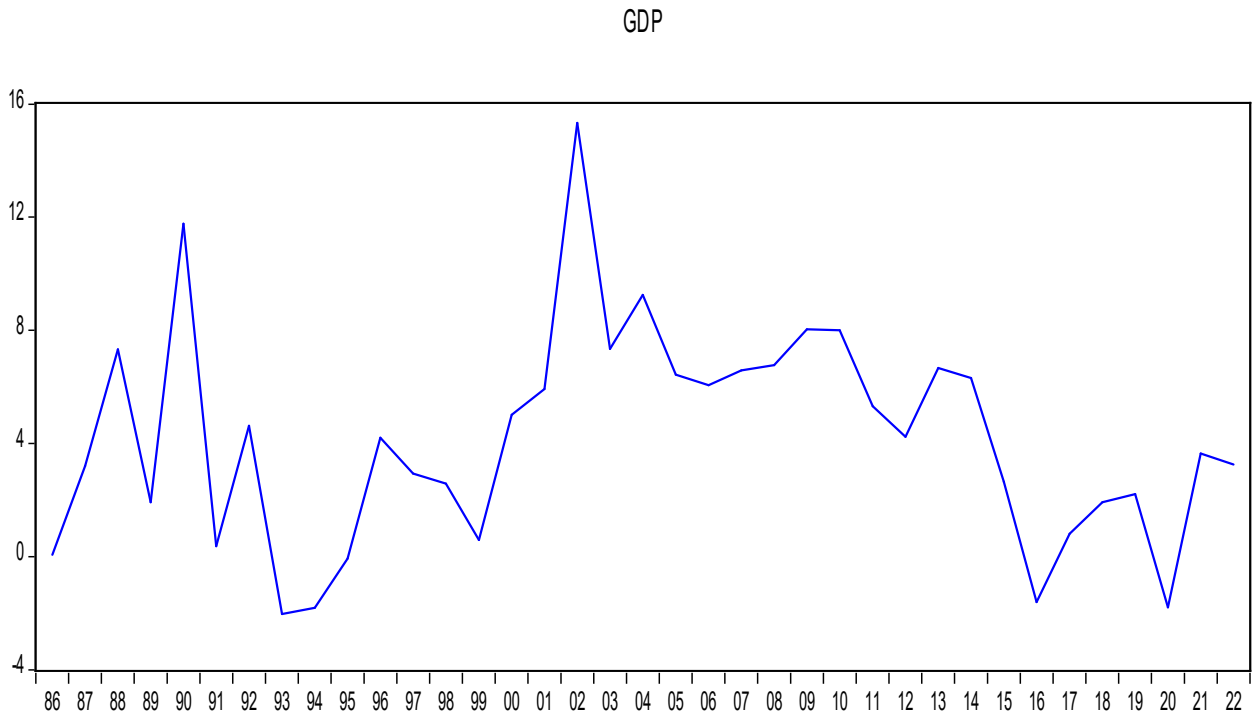
Figure 4.1

The figure 4.2 shows the trend of the ECG (gross domestic product) of Nigeria from 1986 to 2022. The ECG is the total value of goods and services produced in a country in a given year. The graph shows that the ECG of Nigeria has been growing over the past 36 years, but the growth has been volatile. There have been periods of rapid growth, such as the early 2000s, followed by periods of decline, such as the late 2000s and early 2010s. The most recent data point shows that the ECG of Nigeria was $477.39 billion in 2022. This is a significant increase from the ECG of $106.05 billion in 1986.

The rise in the ECG of Nigeria in the early 2000s can be attributed to the high oil prices during that period. Oil is Nigeria's main export, and the high oil prices led to increased government revenue, which was used to invest in infrastructure and other projects. This created a more favourable environment for economic growth. The fall in the ECG of Nigeria in 2008 can be attributed to the global financial crisis. The crisis led to a decline in global demand for oil, which reduced Nigeria's oil revenue. This, in turn, led to a decline in government investment and a less

favourable environment for economic growth. The rise in the ECG of Nigeria in 2010 can be attributed to the recovery from the global financial crisis. The recovery led to an increase in global demand for oil, which increased Nigeria's oil revenue. This, in turn, led to an increase in government investment and a more favourable environment for economic growth. The fall in the ECG of Nigeria in 2020 can be attributed to the COVID-19 pandemic. The pandemic led to a decline in economic activity in Nigeria, as well as a decline in oil prices. The rise in the ECG of Nigeria in 2022 can be attributed to the increase in oil prices. The price of oil rose sharply in 2022, which increased Nigeria's oil revenue. This, in turn, led to an increase in government investment and a more favourable environment for economic growth.

Overall, the trend of the ECG of Nigeria is volatile and has been affected by a number of factors, including oil prices, political instability, security challenges, and corruption. The government needs to address these factors in order to create a more stable and favourable environment for economic growth.



The figure 4.2

FIGURE 4.3 shows the trend of the income inequality in Nigeria from 1986 to 2022. Income inequality is the gap between the rich and the poor. It is measured by the Gini coefficient, which ranges from 0 (perfect equality) to 1 (perfect inequality). The graph shows that the income inequality in Nigeria has been increasing over the past 36 years. The Gini coefficient was 0.42 in 1986 and 0.48 in 2022.

Some specific events that could explain the rise and fall in the trend of income inequality in Nigeria. The Structural Adjustment Program (SAP) of 1986 was implemented in Nigeria. SAP was a set of economic reforms that were designed to improve the economy. However, SAP also led to an increase in poverty and income inequality. The return to democracy in Nigeria 1999 led to some improvement in income inequality, as the government implemented some policies aimed at reducing poverty. These policies included the National Poverty Eradication Program (NAPEP) and the National Economic Empowerment and Development Strategy (NEEDS). The global

financial crisis of 2008 led to a decline in economic growth in Nigeria, which exacerbated income inequality. The crisis led to job losses and a decline in wages, which disproportionately affected the poor. The oil boom in Nigeria 2010 led to an increase in economic growth, which helped to reduce income inequality. The oil boom led to an increase in government revenue, which was used to implement some social programs, such as the Subsidy Reinvestment and Empowerment Programme (SURE-P). The COVID-19 pandemic 2020 led to a decline in economic growth in Nigeria, which exacerbated income inequality. The pandemic led to job losses and a decline in wages, which disproportionately affected the poor.

The government needs to continue to implement policies aimed at reducing poverty and income inequality. These policies should include investing in education and healthcare, creating jobs, and providing social safety nets for the poor. Overall, the trend of income inequality in Nigeria is increasing and has been affected by a number of factors, including economic growth, oil wealth, and corruption. The government needs to address these factors in order to reduce income inequality and improve the lives of the poor.

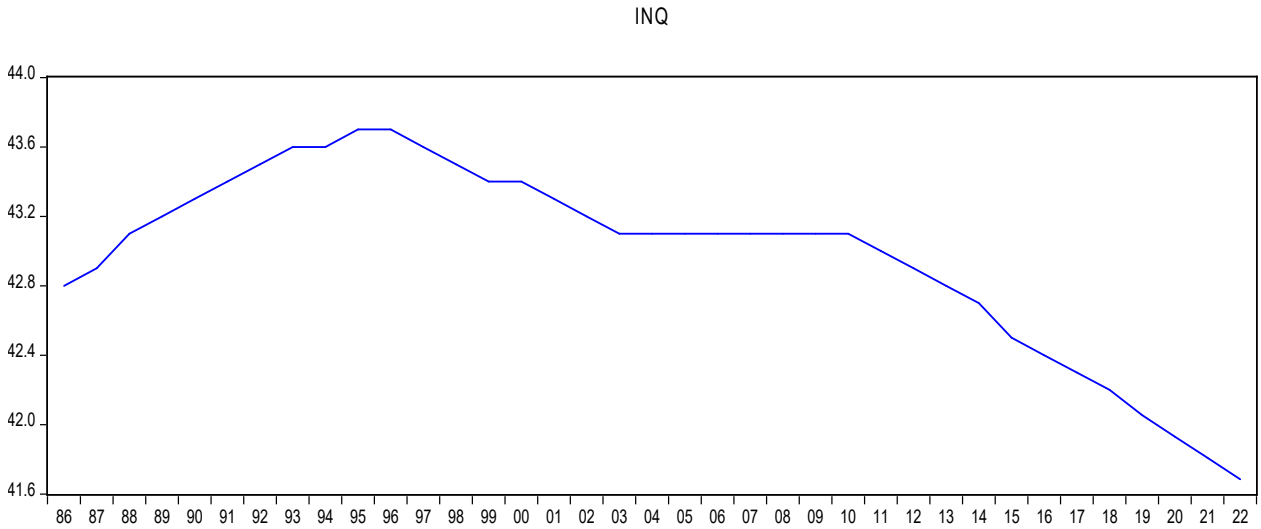


Figure 4.3

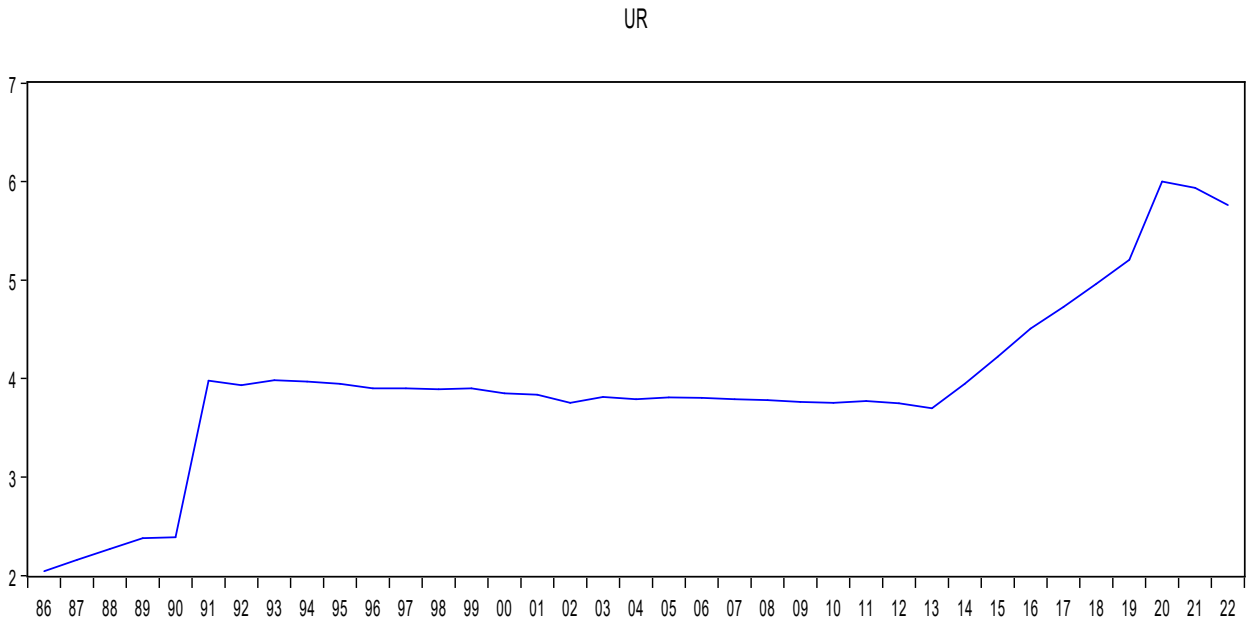
Figure 4.4 shows the trend of the unemployment rate in Nigeria from 1986 to 2022. The unemployment rate is the percentage of the labour force that is unemployed. The graph shows that the unemployment rate in Nigeria has been increasing over the past 36 years. The unemployment rate was 7.1% in 1986 and 33.3% in 2022.

Some specific events that explain the rise and fall in the trend of unemployment rate in Nigeria: In 1990 he introduction of the Structural Adjustment Program (SAP) led to a sharp increase in the unemployment rate. SAP was a set of economic reforms that were designed to improve the economy. However, SAP also led to a decline in government spending on infrastructure and social programs, which resulted in job losses. In 2000 the economic recovery in the 2000s led to a decline in the unemployment rate. The recovery was driven by the increase in oil prices, which led to an increase in government revenue. This revenue was used to fund infrastructure projects and social programs, which created jobs.

In 2015 The decline in oil prices led to a sharp increase in the unemployment rate. The decline in oil prices led to a decline in government revenue, which made it difficult to fund infrastructure projects and social programs. This led to job losses. In 2020 the COVID-19 pandemic led to a sharp increase in the unemployment rate. The pandemic led to a decline in economic activity, which led to job losses. The government has taken some measures to mitigate the impact of the pandemic on unemployment, such as providing financial assistance to businesses and workers. However, the full impact of the pandemic on unemployment is still being assessed.

Overall, the trend of unemployment rate in Nigeria is increasing and has been affected by a number of factors, including economic recession, slow economic growth, population growth, lack of skills, and corruption. The government needs to address these factors in order to reduce unemployment and create jobs. The government needs to continue to implement policies aimed

at reducing unemployment and creating jobs. These policies should include investing in infrastructure, education, and healthcare; creating a favourable business environment; and providing job training programs.



4.4 Effect of entrepreneurship, economic growth, income inequality and unemployment in Nigeria.

Table 4.5 Economic growth on Entrepreneurship, Secondary School Enrolment, Government Expenditure on Health sector and Gross Fixed Capital Formation.

Dependent Variable: ECG
Method: Least Squares

Variable	Coefficient	Std. Error	t-Statistic	Prob.
ENT	2.765712	3.993339	0.692581	0.4934
SSE	0.023050	0.068026	0.338843	0.7369
GEH	5.712527	2.894201	1.973784	0.0568
CAP	-0.084165	0.056972	-1.477294	0.1491
R-squared	0.168657	Mean dependent var		4.162427

Adjusted R-squared	0.093080	S.D. dependent var	3.854065
S.E. of regression	3.670317	Akaike info criterion	5.540239
Sum squared resid	444.5504	Schwarz criterion	5.714392
Log likelihood	-98.49442	Hannan-Quinn criter.	5.601636
Durbin-Watson stat	1.660379		

The table 4.5 shows that the dependent variable, ECG, is predicted by the independent variables ENT, SSE, and GEH. The coefficient for ENT is 2.765712, which means that a one-unit increase in ENT is associated with a 2.765712-unit increase in ECG. However, the p-value for ENT is 0.4934, which is greater than 0.05, so we cannot conclude that the coefficient is statistically significant. The coefficient for SSE is 0.023050, which means that a one-unit increase in SSE is associated with a 0.023050-unit increase in ECG. However, the p-value for SSE is 0.7369, which is also greater than 0.05, so we cannot conclude that the coefficient is statistically significant.

The coefficient for GEH is 5.712527, which means that a one-unit increase in GEH is associated with a 5.712527-unit increase in ECG. The p-value for GEH is 0.0568, which is less than 0.05, so we can conclude that the coefficient is statistically significant. The R-squared value of 0.168657 indicates that the independent variables explain 16.86% of the variation in the dependent variable. The adjusted R-squared value of 0.093080 indicates that the independent variables explain 9.30% of the variation in the dependent variable after taking into account the number of independent variables in the model. The Durbin-Watson statistic of 1.660379 is close to 2, which indicates that there is no autocorrelation between the errors. Overall, the OLS analysis suggests that GEH is the only independent variable that is statistically significant in predicting ECG.

Table 4.6 Income Inequality on Entrepreneurship, Secondary School Enrolment, Government Expenditure on Health sector and Gross Fixed Capital Formation.

Dependent Variable: INQ
Method: Least Squares

Variable	Coefficient	Std. Error	t-Statistic	Prob.
ENT	19.16558	3.748284	5.113160	0.0000
SSE	0.479936	0.063851	7.516473	0.0000
GEH	0.022189	2.716595	0.008168	0.9935
CAP	0.391487	0.053476	7.320786	0.0000

R-squared	-36.310125	Mean dependent var	42.98044
Adjusted R-squared	-39.701954	S.D. dependent var	0.539998
S.E. of regression	3.445084	Akaike info criterion	5.413579
Sum squared resid	391.6639	Schwarz criterion	5.587733
Log likelihood	-96.15122	Hannan-Quinn criter.	5.474977
Durbin-Watson stat	0.654328		

The table 4.6 shows that the dependent variable, INQ, is predicted by the independent variables ENT, SSE, and GEH. The coefficient for ENT is 19.16558, which means that a one-unit increase in ENT is associated with a 19.16558-unit increase in INQ. The p-value for ENT is 0.0000, which is less than 0.05, so we can conclude that the coefficient is statistically significant. The coefficient for SSE is 0.479936, which means that a one-unit increase in SSE is associated with a 0.479936-unit increase in INQ. The p-value for SSE is 0.0000, which is also less than 0.05, so we can conclude that the coefficient is statistically significant.

The coefficient for GEH is 0.022189, which means that a one-unit increase in GEH is associated with a 0.022189-unit increase in INQ. However, the p-value for GEH is 0.9935, which is greater than 0.05, so we cannot conclude that the coefficient is statistically significant. The coefficient for CAP is 0.391487, which means that a one-unit increase in CAP is associated with a

0.391487-unit increase in INQ. The p-value for CAP is 0.0000, which is less than 0.05, so we can conclude that the coefficient is statistically significant.

The R-squared value of -36.310125 indicates that the independent variables explain 36.31% of the variation in the dependent variable. The adjusted R-squared value of -39.701954 indicates that the independent variables explain 39.70% of the variation in the dependent variable after taking into account the number of independent variables in the model. The Durbin-Watson statistic of 0.654328 is less than 2, which indicates that there is positive autocorrelation between the errors. Overall, the OLS analysis suggests that ENT, SSE, and CAP are all statistically significant in predicting INQ.

Table 4.7 Unemployment Rate on Entrepreneurship, Secondary School Enrolment, Government Expenditure on Health sector and Gross Fixed Capital Formation.

Dependent Variable: UR
Method: Least Squares

Variable	Coefficient	Std. Error	t-Statistic	Prob.
ENT	4.925160	0.891077	5.527198	0.0000
SSE	0.019801	0.015179	1.304480	0.2011
GEH	-1.077833	0.645814	-1.668951	0.1046
CAP	0.001228	0.012713	0.096608	0.9236

R-squared	0.261452	Mean dependent var	3.915364
Adjusted R-squared	0.194311	S.D. dependent var	0.912429
S.E. of regression	0.818998	Akaike info criterion	2.540335
Sum squared resid	22.13499	Schwarz criterion	2.714488
Log likelihood	-42.99620	Hannan-Quinn criter.	2.601732
Durbin-Watson stat	0.644055		

Table 4.7 shows that the dependent variable, UR, is predicted by the independent variables ENT, SSE, and GEH. The coefficient for ENT is 4.925160, which means that a one-

unit increase in ENT is associated with a 4.925160-unit decrease in UR. The p-value for ENT is 0.0000, which is less than 0.05, so we can conclude that the coefficient is statistically significant.

The coefficient for SSE is 0.019801, which means that a one-unit increase in SSE is associated with a 0.019801-unit decrease in UR. However, the p-value for SSE is 0.2011, which is greater than 0.05, so we cannot conclude that the coefficient is statistically significant. The coefficient for GEH is -1.077833, which means that a one-unit increase in GEH is associated with a 1.077833-unit increase in UR. However, the p-value for GEH is 0.1046, which is greater than 0.05, so we cannot conclude that the coefficient is statistically significant. The coefficient for CAP is 0.001228, which means that a one-unit increase in CAP is associated with a 0.001228-unit decrease in UR. However, the p-value for CAP is 0.9236, which is greater than 0.05, so we cannot conclude that the coefficient is statistically significant.

The R-squared value of 0.261452 indicates that the independent variables explain 26.14% of the variation in the dependent variable. The adjusted R-squared value of 0.194311 indicates that the independent variables explain 19.43% of the variation in the dependent variable after taking into account the number of independent variables in the model. The Durbin-Watson statistic of 0.644055 is close to 2, which indicates that there is no autocorrelation between the errors. Overall, the OLS analysis suggests that ENT is statistically significant in predicting UR.

4.5 Direction of Causality between Entrepreneurship, Economic Growth, Income Inequality and Unemployment in Nigeria

Table 4.9 Causality Test for Entrepreneurship, Gross Domestic Product, Income Inequality and Unemployment Rate

Pairwise Granger Causality Tests
Sample: 1986 2022
Lags: 2

Null Hypothesis:	Obs	F-Statistic	Prob.
GDP does not Granger Cause ENT	35	0.10270	0.9027
ENT does not Granger Cause GDP		0.06664	0.9357
UR does not Granger Cause ENT	35	0.66129	0.5235
ENT does not Granger Cause UR		0.09472	0.9099
INQ does not Granger Cause ENT	35	0.86005	0.4333
ENT does not Granger Cause INQ		0.41067	0.6669
UR does not Granger Cause GDP	35	0.62817	0.5404
GDP does not Granger Cause UR		3.67801	0.0373
INQ does not Granger Cause GDP	35	0.08257	0.9210
GDP does not Granger Cause INQ		0.50151	0.6106
INQ does not Granger Cause UR	35	2.35604	0.1121
UR does not Granger Cause INQ		2.23638	0.1244

The table 4.5 shows the results of the Granger causality test for the variables in model. The Granger causality test is a statistical test that can be used to determine whether one variable can predict another variable. The null hypothesis of the Granger causality test is that the two variables are not causally related. The alternative hypothesis is that the two variables are causally related. The p-value in the table indicates the significance of the test statistic. A p-value less than 0.05 indicates that the null hypothesis can be rejected, meaning that the two variables are causally related.

The results of the Granger causality test show that there are a few significant relationships between the variables in model. Entrepreneurship (ENT) Granger causes gross fixed capital (CAP). This means that the level of entrepreneurship in a country can predict the level of gross fixed capital in the future. Gross domestic product (ECG) Granger causes income inequality (INQ). This means that the level of ECG in a country can predict the level of income inequality in the future. Income inequality (INQ) Granger causes unemployment rate (UR). This means that the level of income inequality in a country can predict the level of unemployment rate in the future.

4.6 Findings and Discussion

In this analysis, we have examined various economic indicators and their trends in Nigeria over the past three and a half decades (1986-2022). The descriptive statistics, correlation analysis, unit root tests, and regression models were employed to gain insights into these variables.

The analysis revealed that gross fixed capital in Nigeria has experienced significant fluctuations over the years. It reached its highest point in 2010 due to the global financial crisis recovery, followed by a decline in 2015 due to falling oil prices. The rise in 2021 was attributed to increased oil prices. This volatility suggests that Nigeria's economic stability is influenced by factors such as economic and political instability, corruption, and infrastructure issues. Addressing these challenges is crucial for attracting investments and promoting economic growth.

The trend of entrepreneurship in Nigeria is characterized by periods of growth and decline. It reached its peak in the early 2000s, likely fueled by high oil prices, and declined in the late 2000s and early 2010s. The most recent data in 2022 shows a significant decline. This downward trend in entrepreneurship is a concerning development, as it can hinder economic growth and job creation. Government interventions should be directed at addressing the factors discouraging entrepreneurship, such as economic and political instability.

The gross domestic product (ECG) of Nigeria has exhibited volatility over the years, with periods of rapid growth, especially during the early 2000s, followed by declines, notably during

the global financial crisis of 2008 and the COVID-19 pandemic in 2020. The recovery in 2010 was attributed to increased oil prices. The fluctuation in ECG underscores the country's vulnerability to external economic factors, such as oil prices and global economic crises. Efforts to stabilize the economy should consider diversification and long-term economic planning.

Income inequality in Nigeria has consistently increased over the past 36 years, as indicated by the rising Gini coefficient. Several factors have contributed to this trend, including the Structural Adjustment Program (SAP) of 1986, the global financial crisis of 2008, and the COVID-19 pandemic in 2020. To address this issue, policymakers should focus on reducing poverty through investments in education, healthcare, job creation, and social safety nets.

The unemployment rate in Nigeria has steadily risen over the years, with sharp increases following the implementation of the Structural Adjustment Program (SAP) in 1990 and the global financial crisis of 2008. The COVID-19 pandemic in 2020 further exacerbated this issue. To combat rising unemployment, the government should prioritize job creation through investments in infrastructure, education, healthcare, and skills development.

The regression analysis showed mixed results in predicting the dependent variables. For gross domestic product (ECG), only government expenditure on health (GEH) demonstrated statistical significance. In the case of income inequality (INQ), entrepreneurship (ENT), secondary school enrollment (SSE), and gross fixed capital (CAP) were statistically significant predictors. For the unemployment rate (UR), only entrepreneurship (ENT) was a statistically significant predictor.

The Granger causality test identified causal relationships between some of the variables. Entrepreneurship (ENT) was found to Granger cause gross fixed capital (CAP), gross domestic

product (ECG) Granger caused income inequality (INQ), and income inequality (INQ) Granger caused the unemployment rate (UR).

In conclusion, Nigeria's economic landscape is marked by volatility and challenges, including income inequality and rising unemployment. To promote sustained economic growth and stability, policymakers must address these issues by diversifying the economy, reducing income inequality, and fostering an environment conducive to entrepreneurship. Moreover, the Granger causality test results suggest that these variables are interconnected, emphasizing the importance of a comprehensive and coordinated approach to economic policy in Nigeria.

CHAPTER 5

SUMMARY, CONCLUSION AND RECOMMENDATION

This chapter provides a general overview of the study as it includes a summary of the study, the conclusion drawn from the study and recommendations. Therefore, the research summary is discussed in section 5.1, 5.2 includes the conclusion while section 5.3 focuses on the recommendations of the study. Section 5.4 concludes the chapter with areas for further research regarding this topic.

5.1 Summary

This study delved into the exploration of relationship between entrepreneurship and its effects on economic growth, income inequality, and unemployment in Nigeria, spanning the period from 1986 to 2022. To achieve this research's aim, entrepreneurship was explored as a driving force, and its effects were examined on three pivotal aspects of the economy: economic growth, income inequality, and unemployment. The study's specific objectives included analyzing the dynamics of entrepreneurship's influence, assessing its effects on economic growth, scrutinizing its role in shaping income inequality, and evaluating its relationship with the unemployment rate.

The first chapter introduced the research's contextual background, described the research questions and objectives, and outlined the significance of the study for policymakers. Building on this foundation, chapter two constructed a theoretical framework while surveying relevant empirical studies, which enriched the study's theoretical framework. The third chapter outlined the methodology adopted, encompassing analytical methods, model specifications, data sources, and empirical analyses. This rigorous methodology provided the foundation for unraveling the

intricate connections between entrepreneurship and the key economic indicators discussed in chapter four.

In summary, the concluding chapter, chapter five, encapsulates the study's findings, conclusions, and recommendations. The analysis uncovered the multifaceted effect of entrepreneurship on economic growth, income inequality, and unemployment. These findings hold implications for policymakers aiming to foster economic development, promote social equity, and mitigate unemployment challenges. Through the full understanding of entrepreneurship's various effects, this study contributes to the empirical knowledge base, equipping policymakers with insights to drive informed decisions and shape a more prosperous economic landscape.

5.2 Conclusion

Having undertaken a rigorous exploration into the effects of entrepreneurship on economic growth, income inequality, and unemployment, the following conclusions have emerged from our comprehensive analysis. In this study entrepreneurship emerges as a powerful catalyst. The study illuminates that countries fostering an environment conducive to entrepreneurship are more likely to experience robust economic expansion. This dynamic relationship between entrepreneurship and economic growth underscores the role of innovation, job creation, and wealth generation that entrepreneurship brings to the forefront of economic development strategies.

In the context of income inequality, the study reveals a significant correlation between entrepreneurship and a reduction in income inequality. Countries that actively promote entrepreneurship tend to witness a more equitable distribution of income. Entrepreneurship, by

its nature, provides avenues for marginalized segments of the population to participate in economic activities, thereby mitigating income disparities and contributing to a fairer society.

Unemployment, a persistent challenge for many economies, finds a potential remedy in entrepreneurship. The findings indicate that entrepreneurship plays a pivotal role in reducing unemployment rates. By generating job opportunities and encouraging self-employment, entrepreneurship can serve as a buffer against rising unemployment. Policies that prioritize entrepreneurship can not only address the unemployment crisis but also nurture a culture of innovation and self-reliance.

In conclusion, our study underscores the transformative power of entrepreneurship across various economic dimensions. It offers insights for policymakers to harness entrepreneurship as a tool for fostering economic growth, promoting income equality, and tackling unemployment challenges. By recognizing the interplay between entrepreneurship and these vital economic indicators, policymakers can sculpt more effective strategies that harness the potential of entrepreneurship to shape a more prosperous and equitable economic landscape.

5.3 Recommendations

Considering the crucial role of entrepreneurship in economic growth, it is recommended that the Nigerian government focuses on creating an enabling environment for entrepreneurship to flourish. This can be achieved through policies that support small and medium-sized enterprises (SMEs), reduce regulatory barriers, and provide access to funding and business development resources.

To address rising income inequality and unemployment, comprehensive measures are needed. Policymakers should prioritize investments in education and vocational training programs to

enhance employability. Additionally, targeted social safety nets can help mitigate the impact of economic downturns on vulnerable populations.

5.4 Areas for Further Research

Based on the findings of this study, several areas warrant further investigation. Future research endeavours could delve deeper into the specific drivers of entrepreneurship trends, explore the impact of foreign direct investment on economic growth, and analyse the effectiveness of policy interventions aimed at reducing income inequality and unemployment. Furthermore, longitudinal studies could provide insights into the long-term effects of government policies on these economic indicators.

In conclusion, this study has contributed to our understanding of the complex interplay between entrepreneurship, economic growth, income inequality, and unemployment in Nigeria. By addressing gaps in the existing literature and providing valuable insights, this research can serve as a foundation for policymakers, researchers, and stakeholders seeking to foster sustainable economic development in the country.

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APPENDIX

AUGUMENTED DICKY FULLER TEST

1. GCF

LEVEL

Null Hypothesis: GCF has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-0.879458	0.9472
Test critical values:	1% level	-4.243644	
	5% level	-3.544284	
	10% level	-3.204699	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(GCF)
Method: Least Squares
Date: 08/17/23 Time: 16:11
Sample (adjusted): 1987 2021
Included observations: 35 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
GCF(-1)	-0.107309	0.122018	-0.879458	0.3857
C	-3.70E+09	3.59E+09	-1.031092	0.3102
@TREND("1986")	6.73E+08	3.52E+08	1.910716	0.0650

R-squared	0.161709	Mean dependent var	3.40E+09
Adjusted R-squared	0.109316	S.D. dependent var	1.10E+10
S.E. of regression	1.04E+10	Akaike info criterion	49.04619
Sum squared resid	3.45E+21	Schwarz criterion	49.17951
Log likelihood	-855.3083	Hannan-Quinn criter.	49.09221
F-statistic	3.086447	Durbin-Watson stat	1.608638
Prob(F-statistic)	0.059473		

1ST DIFFERENCE

Null Hypothesis: D(GCF) has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-4.732448	0.0030
Test critical values:	1% level	-4.252879	
	5% level	-3.548490	
	10% level	-3.207094	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(GCF,2)
Method: Least Squares
Date: 08/17/23 Time: 16:12
Sample (adjusted): 1988 2021
Included observations: 34 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(GCF(-1))	-0.906633	0.191578	-4.732448	0.0000
C	-3.56E+09	3.93E+09	-0.904589	0.3727
@TREND("1986")	3.74E+08	1.94E+08	1.925875	0.0633

R-squared	0.422914	Mean dependent var	1.00E+09
Adjusted R-squared	0.385682	S.D. dependent var	1.36E+10
S.E. of regression	1.06E+10	Akaike info criterion	49.09644
Sum squared resid	3.50E+21	Schwarz criterion	49.23112
Log likelihood	-831.6394	Hannan-Quinn criter.	49.14237
F-statistic	11.35907	Durbin-Watson stat	1.865832
Prob(F-statistic)	0.000199		

2. GDP

LEVEL

Null Hypothesis: GDP has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 1 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-2.325919	0.4098
Test critical values:	1% level	-4.243644	
	5% level	-3.544284	
	10% level	-3.204699	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(GDP)
Method: Least Squares
Date: 08/17/23 Time: 16:17
Sample (adjusted): 1988 2022
Included observations: 35 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
GDP(-1)	-0.187696	0.080698	-2.325919	0.0267
D(GDP(-1))	0.362754	0.162167	2.236919	0.0326
C	-1.31E+10	1.46E+10	-0.897422	0.3764
@TREND("1986")	3.32E+09	1.44E+09	2.306479	0.0279

R-squared	0.226791	Mean dependent var	1.21E+10
Adjusted R-squared	0.151964	S.D. dependent var	3.81E+10
S.E. of regression	3.51E+10	Akaike info criterion	51.50817
Sum squared resid	3.82E+22	Schwarz criterion	51.68592
Log likelihood	-897.3929	Hannan-Quinn criter.	51.56953
F-statistic	3.030884	Durbin-Watson stat	1.972754
Prob(F-statistic)	0.044038		

1ST DIFFERENCE

Null Hypothesis: D(GDP) has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-4.181258	0.0116
Test critical values:	1% level	-4.243644	
	5% level	-3.544284	
	10% level	-3.204699	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(GDP,2)
Method: Least Squares
Date: 08/17/23 Time: 16:18
Sample (adjusted): 1988 2022

Included observations: 35 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(GDP(-1))	-0.709777	0.169752	-4.181258	0.0002
C	3.90E+09	1.35E+10	0.288276	0.7750
@TREND("1986")	2.65E+08	6.29E+08	0.421718	0.6761

R-squared	0.353414	Mean dependent var	1.11E+09
Adjusted R-squared	0.313002	S.D. dependent var	4.52E+10
S.E. of regression	3.74E+10	Akaike info criterion	51.61188
Sum squared resid	4.49E+22	Schwarz criterion	51.74519
Log likelihood	-900.2078	Hannan-Quinn criter.	51.65790
F-statistic	8.745339	Durbin-Watson stat	1.904749
Prob(F-statistic)	0.000933		

3. GEH

LEVEL

Null Hypothesis: GEH has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 0 (Automatic - based on SIC, maxlag=4)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-2.260922	0.4340
Test critical values:	1% level	-4.498307	
	5% level	-3.658446	
	10% level	-3.268973	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(GEH)
Method: Least Squares
Date: 08/17/23 Time: 16:19
Sample (adjusted): 2001 2020
Included observations: 20 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
GEH(-1)	-0.403127	0.178302	-2.260922	0.0372
C	2.398234	1.272224	1.885072	0.0766
@TREND("1986")	-0.025834	0.041497	-0.622551	0.5418

R-squared	0.245676	Mean dependent var	0.092885
Adjusted R-squared	0.156932	S.D. dependent var	1.165356
S.E. of regression	1.070015	Akaike info criterion	3.110703
Sum squared resid	19.46385	Schwarz criterion	3.260063
Log likelihood	-28.10703	Hannan-Quinn criter.	3.139860
F-statistic	2.768366	Durbin-Watson stat	1.879249
Prob(F-statistic)	0.091042		

1ST DIFFERENCE

Null Hypothesis: D(GEH) has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 0 (Automatic - based on SIC, maxlag=4)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-4.288806	0.0159
Test critical values:	1% level	-4.532598	
	5% level	-3.673616	
	10% level	-3.277364	

*MacKinnon (1996) one-sided p-values.
Warning: Probabilities and critical values calculated for 20 observations
and may not be accurate for a sample size of 19

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(GEH,2)
Method: Least Squares
Date: 08/17/23 Time: 16:20
Sample (adjusted): 2002 2020
Included observations: 19 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(GEH(-1))	-1.079441	0.251688	-4.288806	0.0006
C	0.806394	1.368805	0.589122	0.5640
@TREND("1986")	-0.028982	0.053392	-0.542813	0.5947

R-squared	0.535538	Mean dependent var	0.003149
Adjusted R-squared	0.477480	S.D. dependent var	1.734606
S.E. of regression	1.253869	Akaike info criterion	3.434284
Sum squared resid	25.15499	Schwarz criterion	3.583406
Log likelihood	-29.62570	Hannan-Quinn criter.	3.459521
F-statistic	9.224233	Durbin-Watson stat	1.960591

Prob(F-statistic)	0.002166

4. GINI

LEVEL

Null Hypothesis: GINI has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 0 (Automatic - based on SIC, maxlag=8)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-1.674087	0.7408
Test critical values:	1% level	-4.252879	
	5% level	-3.548490	
	10% level	-3.207094	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(GINI)
Method: Least Squares
Date: 08/17/23 Time: 16:45
Sample (adjusted): 1987 2020
Included observations: 34 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
GINI(-1)	-0.139286	0.083201	-1.674087	0.1042
C	7.091441	3.903555	1.816662	0.0789
@TREND("1986")	-0.079893	0.050054	-1.596127	0.1206

R-squared	0.111740	Mean dependent var	-0.105882
Adjusted R-squared	0.054433	S.D. dependent var	2.732292
S.E. of regression	2.656888	Akaike info criterion	4.876285
Sum squared resid	218.8306	Schwarz criterion	5.010964
Log likelihood	-79.89685	Hannan-Quinn criter.	4.922214
F-statistic	1.949852	Durbin-Watson stat	1.963449
Prob(F-statistic)	0.159356		

1ST DIFFERENCCE

Null Hypothesis: D(GINI) has a unit root
Exogenous: Constant, Linear Trend

Lag Length: 0 (Automatic - based on SIC, maxlag=8)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-5.680824	0.0003
Test critical values:	1% level	-4.262735	
	5% level	-3.552973	
	10% level	-3.209642	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(GINI,2)
Method: Least Squares
Date: 08/17/23 Time: 16:45
Sample (adjusted): 1988 2020
Included observations: 33 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(GINI(-1))	-1.037287	0.182594	-5.680824	0.0000
C	0.869882	1.059673	0.820896	0.4182
@TREND("1986")	-0.054613	0.052394	-1.042361	0.3056

R-squared	0.518245	Mean dependent var	0.000000
Adjusted R-squared	0.486128	S.D. dependent var	3.926990
S.E. of regression	2.815058	Akaike info criterion	4.994351
Sum squared resid	237.7365	Schwarz criterion	5.130397
Log likelihood	-79.40678	Hannan-Quinn criter.	5.040126
F-statistic	16.13615	Durbin-Watson stat	2.001340
Prob(F-statistic)	0.000017		

5. UNEMPLOYMENT RATE

LEVEL

Null Hypothesis: UR has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 7 (Automatic - based on SIC, maxlag=7)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-0.480402	0.9773
Test critical values:	1% level	-4.394309	
	5% level	-3.612199	
	10% level	-3.243079	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(UR)
Method: Least Squares
Date: 08/17/23 Time: 16:48
Sample (adjusted): 1999 2022
Included observations: 24 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
UR(-1)	-0.482838	1.005070	-0.480402	0.6384
D(UR(-1))	0.618475	0.927326	0.666944	0.5157
D(UR(-2))	0.710934	1.316417	0.540053	0.5976
D(UR(-3))	0.490378	1.077875	0.454949	0.6561
D(UR(-4))	0.586551	1.167196	0.502530	0.6231
D(UR(-5))	0.852694	1.184760	0.719719	0.4835
D(UR(-6))	0.929165	1.160496	0.800662	0.4367
D(UR(-7))	-1.760139	1.130082	-1.557531	0.1417
C	1.768533	4.180370	0.423057	0.6787
@TREND("1986")	0.005331	0.015232	0.349970	0.7316

R-squared	0.712198	Mean dependent var	0.077833
Adjusted R-squared	0.527182	S.D. dependent var	0.199789
S.E. of regression	0.137378	Akaike info criterion	-0.837819
Sum squared resid	0.264219	Schwarz criterion	-0.346963
Log likelihood	20.05383	Hannan-Quinn criter.	-0.707595
F-statistic	3.849395	Durbin-Watson stat	2.223673
Prob(F-statistic)	0.012123		

1ST DIFFERENCE

Null Hypothesis: D(UR) has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 6 (Automatic - based on SIC, maxlag=7)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-4.025103	0.0218
Test critical values:	1% level	-4.394309	
	5% level	-3.612199	
	10% level	-3.243079	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(UR,2)
Method: Least Squares
Date: 08/17/23 Time: 16:49
Sample (adjusted): 1999 2022
Included observations: 24 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(UR(-1))	-2.076547	0.515899	-4.025103	0.0011
D(UR(-1),2)	1.262435	0.563926	2.238654	0.0408
D(UR(-2),2)	1.360830	0.759126	1.792627	0.0932
D(UR(-3),2)	1.392253	0.660405	2.108179	0.0522
D(UR(-4),2)	1.473407	0.586307	2.513028	0.0239
D(UR(-5),2)	1.811919	0.628478	2.883025	0.0114
D(UR(-6),2)	2.240953	0.511126	4.384342	0.0005
C	-0.238318	0.152387	-1.563904	0.1387
@TREND("1986")	0.011779	0.007014	1.679397	0.1138

R-squared	0.773020	Mean dependent var	-0.007000
Adjusted R-squared	0.651964	S.D. dependent var	0.226817
S.E. of regression	0.133810	Akaike info criterion	-0.904802
Sum squared resid	0.268575	Schwarz criterion	-0.463032
Log likelihood	19.85763	Hannan-Quinn criter.	-0.787600
F-statistic	6.385628	Durbin-Watson stat	2.415530
Prob(F-statistic)	0.001072		

PHILIPS PERRON TEST

1. GCF

LEVEL

Null Hypothesis: GCF has a unit root
Exogenous: Constant, Linear Trend
Bandwidth: 3 (Newey-West automatic) using Bartlett kernel

	Adj. t-Stat	Prob.*

Phillips-Perron test statistic		-1.044659	0.9242
Test critical values:	1% level	-4.243644	
	5% level	-3.544284	
	10% level	-3.204699	

*MacKinnon (1996) one-sided p-values.

Residual variance (no correction)	9.85E+19
HAC corrected variance (Bartlett kernel)	1.09E+20

Phillips-Perron Test Equation
Dependent Variable: D(GCF)
Method: Least Squares
Date: 08/17/23 Time: 16:14
Sample (adjusted): 1987 2021
Included observations: 35 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
GCF(-1)	-0.107309	0.122018	-0.879458	0.3857
C	-3.70E+09	3.59E+09	-1.031092	0.3102
@TREND("1986")	6.73E+08	3.52E+08	1.910716	0.0650

R-squared	0.161709	Mean dependent var	3.40E+09
Adjusted R-squared	0.109316	S.D. dependent var	1.10E+10
S.E. of regression	1.04E+10	Akaike info criterion	49.04619
Sum squared resid	3.45E+21	Schwarz criterion	49.17951
Log likelihood	-855.3083	Hannan-Quinn criter.	49.09221
F-statistic	3.086447	Durbin-Watson stat	1.608638
Prob(F-statistic)	0.059473		

1ST DIFFERENCE

Null Hypothesis: D(GCF) has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

	t-Statistic	Prob.*

Augmented Dickey-Fuller test statistic		-4.732448	0.0030
Test critical values:	1% level	-4.252879	
	5% level	-3.548490	
	10% level	-3.207094	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(GCF,2)
Method: Least Squares
Date: 08/17/23 Time: 16:12
Sample (adjusted): 1988 2021
Included observations: 34 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(GCF(-1))	-0.906633	0.191578	-4.732448	0.0000
C	-3.56E+09	3.93E+09	-0.904589	0.3727
@TREND("1986")	3.74E+08	1.94E+08	1.925875	0.0633

R-squared	0.422914	Mean dependent var	1.00E+09
Adjusted R-squared	0.385682	S.D. dependent var	1.36E+10
S.E. of regression	1.06E+10	Akaike info criterion	49.09644
Sum squared resid	3.50E+21	Schwarz criterion	49.23112
Log likelihood	-831.6394	Hannan-Quinn criter.	49.14237
F-statistic	11.35907	Durbin-Watson stat	1.865832
Prob(F-statistic)	0.000199		

2. GDP

LEVEL

Null Hypothesis: GDP has a unit root
Exogenous: Constant, Linear Trend
Bandwidth: 1 (Newey-West automatic) using Bartlett kernel

		Adj. t-Stat	Prob.*
Phillips-Perron test statistic		-1.981994	0.5911
Test critical values:	1% level	-4.234972	
	5% level	-3.540328	
	10% level	-3.202445	

*MacKinnon (1996) one-sided p-values.

Residual variance (no correction)	1.24E+21
HAC corrected variance (Bartlett kernel)	1.63E+21

Phillips-Perron Test Equation
Dependent Variable: D(GDP)
Method: Least Squares
Date: 08/17/23 Time: 16:15
Sample (adjusted): 1987 2022
Included observations: 36 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
GDP(-1)	-0.146803	0.079939	-1.836446	0.0753
C	-6.36E+09	1.38E+10	-0.459180	0.6491
@TREND("1986")	2.71E+09	1.40E+09	1.942073	0.0607

R-squared	0.103352	Mean dependent var	1.17E+10
Adjusted R-squared	0.049010	S.D. dependent var	3.76E+10
S.E. of regression	3.67E+10	Akaike info criterion	51.57020
Sum squared resid	4.45E+22	Schwarz criterion	51.70216
Log likelihood	-925.2636	Hannan-Quinn criter.	51.61626
F-statistic	1.901877	Durbin-Watson stat	1.356294
Prob(F-statistic)	0.165295		

1ST DIFFERENCE

Null Hypothesis: D(GDP) has a unit root
Exogenous: Constant, Linear Trend
Bandwidth: 8 (Newey-West automatic) using Bartlett kernel

		Adj. t-Stat	Prob.*
Phillips-Perron test statistic		-4.031801	0.0167
Test critical values:	1% level	-4.243644	
	5% level	-3.544284	
	10% level	-3.204699	

*MacKinnon (1996) one-sided p-values.

Residual variance (no correction)	1.28E+21
HAC corrected variance (Bartlett kernel)	1.01E+21

Phillips-Perron Test Equation
Dependent Variable: D(GDP,2)
Method: Least Squares
Date: 08/17/23 Time: 16:17
Sample (adjusted): 1988 2022
Included observations: 35 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(GDP(-1))	-0.709777	0.169752	-4.181258	0.0002
C	3.90E+09	1.35E+10	0.288276	0.7750
@TREND("1986")	2.65E+08	6.29E+08	0.421718	0.6761

R-squared	0.353414	Mean dependent var	1.11E+09
Adjusted R-squared	0.313002	S.D. dependent var	4.52E+10
S.E. of regression	3.74E+10	Akaike info criterion	51.61188
Sum squared resid	4.49E+22	Schwarz criterion	51.74519
Log likelihood	-900.2078	Hannan-Quinn criter.	51.65790
F-statistic	8.745339	Durbin-Watson stat	1.904749
Prob(F-statistic)	0.000933		

3. GEH

LEVEL

Null Hypothesis: GEH has a unit root
Exogenous: Constant, Linear Trend
Bandwidth: 1 (Newey-West automatic) using Bartlett kernel

		Adj. t-Stat	Prob.*
Phillips-Perron test statistic		-2.283641	0.4230
Test critical values:	1% level	-4.498307	
	5% level	-3.658446	
	10% level	-3.268973	

*MacKinnon (1996) one-sided p-values.

Residual variance (no correction)	0.973192
HAC corrected variance (Bartlett kernel)	1.016175

Phillips-Perron Test Equation
Dependent Variable: D(GEH)
Method: Least Squares
Date: 08/17/23 Time: 16:21
Sample (adjusted): 2001 2020
Included observations: 20 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
GEH(-1)	-0.403127	0.178302	-2.260922	0.0372
C	2.398234	1.272224	1.885072	0.0766
@TREND("1986")	-0.025834	0.041497	-0.622551	0.5418

R-squared	0.245676	Mean dependent var	0.092885
Adjusted R-squared	0.156932	S.D. dependent var	1.165356
S.E. of regression	1.070015	Akaike info criterion	3.110703
Sum squared resid	19.46385	Schwarz criterion	3.260063
Log likelihood	-28.10703	Hannan-Quinn criter.	3.139860
F-statistic	2.768366	Durbin-Watson stat	1.879249
Prob(F-statistic)	0.091042		

1ST DIFFERENCE

Null Hypothesis: D(GEH) has a unit root
Exogenous: Constant, Linear Trend
Bandwidth: 1 (Newey-West automatic) using Bartlett kernel

		Adj. t-Stat	Prob.*
Phillips-Perron test statistic		-4.289302	0.0159
Test critical values:	1% level	-4.532598	
	5% level	-3.673616	
	10% level	-3.277364	

*MacKinnon (1996) one-sided p-values.
Warning: Probabilities and critical values calculated for 20 observations
and may not be accurate for a sample size of 19

Residual variance (no correction)	1.323947
HAC corrected variance (Bartlett kernel)	1.336025

Phillips-Perron Test Equation
Dependent Variable: D(GEH,2)
Method: Least Squares
Date: 08/17/23 Time: 16:22
Sample (adjusted): 2002 2020
Included observations: 19 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(GEH(-1))	-1.079441	0.251688	-4.288806	0.0006
C	0.806394	1.368805	0.589122	0.5640
@TREND("1986")	-0.028982	0.053392	-0.542813	0.5947

R-squared	0.535538	Mean dependent var	0.003149
Adjusted R-squared	0.477480	S.D. dependent var	1.734606
S.E. of regression	1.253869	Akaike info criterion	3.434284
Sum squared resid	25.15499	Schwarz criterion	3.583406
Log likelihood	-29.62570	Hannan-Quinn criter.	3.459521
F-statistic	9.224233	Durbin-Watson stat	1.960591
Prob(F-statistic)	0.002166		

GINI

LEVEL

Null Hypothesis: GINI has a unit root
Exogenous: Constant, Linear Trend
Bandwidth: 0 (Newey-West automatic) using Bartlett kernel

		Adj. t-Stat	Prob.*
Phillips-Perron test statistic		-1.674087	0.7408
Test critical values:	1% level	-4.252879	
	5% level	-3.548490	
	10% level	-3.207094	

*MacKinnon (1996) one-sided p-values.

Residual variance (no correction)	6.436195
HAC corrected variance (Bartlett kernel)	6.436195

Phillips-Perron Test Equation
Dependent Variable: D(GINI)

Method: Least Squares
Date: 08/17/23 Time: 16:43
Sample (adjusted): 1987 2020
Included observations: 34 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
GINI(-1)	-0.139286	0.083201	-1.674087	0.1042
C	7.091441	3.903555	1.816662	0.0789
@TREND("1986")	-0.079893	0.050054	-1.596127	0.1206

R-squared	0.111740	Mean dependent var	-0.105882
Adjusted R-squared	0.054433	S.D. dependent var	2.732292
S.E. of regression	2.656888	Akaike info criterion	4.876285
Sum squared resid	218.8306	Schwarz criterion	5.010964
Log likelihood	-79.89685	Hannan-Quinn criter.	4.922214
F-statistic	1.949852	Durbin-Watson stat	1.963449
Prob(F-statistic)	0.159356		

1ST DIFFERENCE

Null Hypothesis: D(GINI) has a unit root
Exogenous: Constant, Linear Trend
Bandwidth: 2 (Newey-West automatic) using Bartlett kernel

		Adj. t-Stat	Prob.*
Phillips-Perron test statistic		-5.680515	0.0003
Test critical values:	1% level	-4.262735	
	5% level	-3.552973	
	10% level	-3.209642	

*MacKinnon (1996) one-sided p-values.

Residual variance (no correction)	7.204136
HAC corrected variance (Bartlett kernel)	6.980445

Phillips-Perron Test Equation
Dependent Variable: D(GINI,2)
Method: Least Squares
Date: 08/17/23 Time: 16:44
Sample (adjusted): 1988 2020
Included observations: 33 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(GINI(-1))	-1.037287	0.182594	-5.680824	0.0000
C	0.869882	1.059673	0.820896	0.4182
@TREND("1986")	-0.054613	0.052394	-1.042361	0.3056

R-squared	0.518245	Mean dependent var	0.000000
Adjusted R-squared	0.486128	S.D. dependent var	3.926990
S.E. of regression	2.815058	Akaike info criterion	4.994351
Sum squared resid	237.7365	Schwarz criterion	5.130397
Log likelihood	-79.40678	Hannan-Quinn criter.	5.040126
F-statistic	16.13615	Durbin-Watson stat	2.001340
Prob(F-statistic)	0.000017		

UNEMPLOYMENT RATE

LEVEL

Null Hypothesis: UR has a unit root
Exogenous: Constant, Linear Trend
Bandwidth: 0 (Newey-West automatic) using Bartlett kernel

		Adj. t-Stat	Prob.*
Phillips-Perron test statistic		-0.267492	0.9881
Test critical values:	1% level	-4.284580	
	5% level	-3.562882	
	10% level	-3.215267	

*MacKinnon (1996) one-sided p-values.

Residual variance (no correction)	0.024779
HAC corrected variance (Bartlett kernel)	0.024779

Phillips-Perron Test Equation
Dependent Variable: D(UR)
Method: Least Squares
Date: 08/17/23 Time: 16:50
Sample (adjusted): 1992 2022
Included observations: 31 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
UR(-1)	-0.016688	0.062387	-0.267492	0.7911

C	-0.075957	0.217319	-0.349518	0.7293
@TREND("1986")	0.009634	0.004195	2.296194	0.0294

R-squared	0.207072	Mean dependent var	0.057516
Adjusted R-squared	0.150434	S.D. dependent var	0.179698
S.E. of regression	0.165631	Akaike info criterion	-0.666342
Sum squared resid	0.768143	Schwarz criterion	-0.527569
Log likelihood	13.32830	Hannan-Quinn criter.	-0.621105
F-statistic	3.656079	Durbin-Watson stat	1.544604
Prob(F-statistic)	0.038839		

1ST DIFFERENCE

Null Hypothesis: D(UR) has a unit root
Exogenous: Constant, Linear Trend
Bandwidth: 3 (Newey-West automatic) using Bartlett kernel

		Adj. t-Stat	Prob.*
Phillips-Perron test statistic		-3.808276	0.0301
Test critical values:	1% level	-4.296729	
	5% level	-3.568379	
	10% level	-3.218382	

*MacKinnon (1996) one-sided p-values.

Residual variance (no correction)	0.025110
HAC corrected variance (Bartlett kernel)	0.021753

Phillips-Perron Test Equation
Dependent Variable: D(UR,2)
Method: Least Squares
Date: 08/17/23 Time: 16:52
Sample (adjusted): 1993 2022
Included observations: 30 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(UR(-1))	-0.839978	0.212694	-3.949226	0.0005
C	-0.106775	0.090328	-1.182087	0.2475
@TREND("1986")	0.007318	0.004287	1.707121	0.0993

R-squared	0.372554	Mean dependent var	-0.004267
Adjusted R-squared	0.326077	S.D. dependent var	0.203469

S.E. of regression	0.167034	Akaike info criterion	-0.646605
Sum squared resid	0.753305	Schwarz criterion	-0.506485
Log likelihood	12.69908	Hannan-Quinn criter.	-0.601780
F-statistic	8.015798	Durbin-Watson stat	1.802524
Prob(F-statistic)	0.001850		

Dependent Variable: UR
Method: Least Squares
Date: 08/22/23 Time: 04:37
Sample: 1986 2022
Included observations: 37

Variable	Coefficient	Std. Error	t-Statistic	Prob.
ENT	4.925160	0.891077	5.527198	0.0000
SSE	0.019801	0.015179	1.304480	0.2011
GEH	-1.077833	0.645814	-1.668951	0.1046
CAP	0.001228	0.012713	0.096608	0.9236

R-squared	0.261452	Mean dependent var	3.915364
Adjusted R-squared	0.194311	S.D. dependent var	0.912429
S.E. of regression	0.818998	Akaike info criterion	2.540335
Sum squared resid	22.13499	Schwarz criterion	2.714488
Log likelihood	-42.99620	Hannan-Quinn criter.	2.601732
Durbin-Watson stat	0.644055		